

SCANNED

AS SENT TO CLASS

21 NOV 1991

STABILITY BOOKLET

m.v. "KRESNA DUA"

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Germanischer Lloyd

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GENERAL PARTICULARS

Ship's Name	:	KRESNA DUA
Official Number	:	384997
Port of Registry	:	SINGAPORE
Owner's Name & Address	:	MALDIVES FISHERY PTE LTD 140, ROBINSON ROAD, #04-04, CHOW HOUSE, SINGAPORE 0106.
Builder's Name & Address	:	GREENBAY MARINE PTE LTD 4, PIONEER SECTOR 1 JURONG SINGAPORE 2262
Hull No	:	P64
Date Keel Laid	:	1990
Principal Dimensions	:	Length Overall = 21.00 Meters Length B.P. = 19.85 Meters Breadth Moulded = 6.40 Meters Depth Moulded = 2.40 Meters
Summer Load Draft	:	1.798 Meters
Block Coefficient	:	0.609
Displacement	:	148.2 (Ext.)
Total Deadweight	:	59.4 (Ext.)
Propulsion Machinery	:	2 x CAT 3408TA 402 BHP @ 1800RPM TWIN DISC MG 516 Gearbox Ratio 4.5:1
Propellers	:	2 x 4-blades; BAR 0.60 Diameter 1300mm x Pitch 1351mm Manganese Bronze

INCLINING EXPERIMENT REPORT

NAME OF VESSEL : "KRESNA DUA"
HULL NO. : P-64
PLACE OF INCLINING : GREENBAY MARINE , SINGAPORE
DATE OF INCLINING : 30th OCTOBER 1991
TIME STARTED : 1400 HOURS
WEATHER : FINE
TIDE : HIGH AND SLACK THROUGHOUT THE TEST
CONDITION OF VESSEL : VIRTUALLY COMPLETE. ALL BILGES AND TANKS DRY.

Present.

MR CHONG , CLASS SURVEYOR
MR J. H. WANG , GREENBAY
MR S. M. TING , GREENBAY
MR Y. K. LAI, GREENBAY
MR FRANCIS, GREENBAY
MR S. K. LOW , GREENBAY
TWO NOS. PERSONNEL

Draughts recorded during experiment

	Port	Starboard	Mean
Aft	1.320	1.375	1.348
Forward	1.320	1.360	1.340

Trim between end recorded draughts 0.007 m in 16.000 m.

Moulded draughts

Mean after draught corrected to aft perp.

$$1.348 + (0.007 / 16.000) * 2.000 = 1.348 \text{ m.}$$

Mean forward draught corrected to forward perp.

$$1.340 - (0.007 / 16.000) * 2.600 = 1.339 \text{ m.}$$

$$\text{Trim in LBP} = 0.010 \text{ m.}$$

$$\text{Mean of LBP draughts} = 1.344 \text{ m.}$$

$$\text{Mean of amidship recorded draughts} = 1.344 \text{ m.}$$

The ship is floating with a straight keel.

Hydrostatics and waterplane particulars as inclined.

the LCG is corrected for trim and where relevant the other figures have been corrected for specific gravity.

Specific gravity of water	=	1.027
Extreme displacement	=	96.731 tonnes
Vertical centre of buoyancy (AB)	=	0.836 metres
Longitudinal centre of buoyancy (AP)	=	9.968 metres
Longitudinal centre of gravity (AP)	=	9.954 metres
Waterplane area	=	104.927 sq metres
Longitudinal centre of flotation (AP)	=	9.730 metres
Tonnes per metre immersion	=	107.762
Moment to change trim one metre	=	135.611 tonne metres
Transverse metacentre above base	=	4.066 metres

Information on pendulums.
-----2 Pendulums, situated as described below.

AFT U-TUBE LOCATED AT FR.4 (2.00M FROM FR.0)

FWD U-TUBE LOCATED AT FR.36 (18.0M FROM FR.0)

Shift:		Weight movements	Moments	Pendulum deflections	
		Weight * distance	T-M	Pendulum 1 MM	Pendulum 2 MM
:	1	1.690 * 4.090	6.912	254.00	207.00
:	2	2.003 * 4.090	8.194	313.00	243.00
:	3	2.003 * 4.090	8.194	301.00	233.00
:	4	1.680 * 4.090	6.871	263.00	202.00
:	5	1.981 * 4.090	8.102	303.00	237.00
:	6	1.981 * 4.090	8.102	303.00	237.00

:	Totals		46.377	1737.00	1359.00

:	Mean pendulum readings			289.50	226.50

:	Mean tangents			0.0513	0.0517

:	Sum of mean tangents				0.1030

Average of mean tangents = 0.0515.

Mean of heeling moments = 7.729 tonne-metres.

G.M. Fluid = $7.729 / (96.731 * 0.0515) = 1.552 \text{ m}$ Free surface correction = $0.000 / 96.731 = + 0.000 \text{ m}$

G.M. SOLID = 1.552 M

VCG solid as inclined (KMT-GM Solid) = $4.066 - 1.552 = 2.514 \text{ m}$

Adjustments to the as Inclined figures.

Items to come off vessel

Item	Weight : Tonnes	Vcg : Metres	V.Mts : T-M	Lcg : Metres	L.Mts : T-M
:PERSONNEL ON BOARD	: 0.520:	3.425:	1.781:	7.313:	3.802:
:CONCRETE BLOCKS	: 3.430:	3.915:	13.428:	17.670:	60.608:
:INCLINING WEIGHTS	: 3.984:	2.950:	11.754:	3.444:	13.723:
: Total to come off	: 7.934:	3.398:	26.964:	9.847:	78.133:

Summary of results.

Item	Weight	Vcg	V.Mts	Lcg	L.Mts
	Tonnes	Metres	T-M	Metres	T-M
Vessel as inclined	96.73	2.514	243.21	9.954	962.91
Items to come off	7.93	3.398	26.96	9.847	78.13
Lightship	88.80	2.435	216.25	9.964	884.78

NOTES TO MASTER

- 1) When loading cargo, attention should be paid to the need for maintaining freeboard at the stern having regard to the safety of the crew working on deck and also to the adverse effect of trim by stern on the stability of the vessel.
- 2) All doors and hatches, etc., opening on to the exposed deck should be kept closed and secured at sea (except for access).
- 3) When cargo is discharged at sea, great care must be taken to maintain the vessel in a stable condition during the discharging operation.
Attention should also be paid to the trimming and heeling of the vessel during unloading.
 - a) It should be noted that when liquid cargo is to be discharged, as soon as pumping commences, a full Free Surface will exist in those tanks being pumped and the effect of this on the stability of the vessel should be taken into account.
 - b) When a tank is to be ballasted at sea to counteract the removal of cargo from the vessel, it should be noted that :
 - i) the tank will have a complete Free surface as soon as ballasting commences and ;
 - ii) that this Free Surface will adversely affect the stability of the vessel before any benefit can be assumed from the weight of ballast water taken on board.

Similar precautions should be adopted when Loading cargo at sea.

- 4) Where Port and Starboard tanks are cross-connected, such connections should be closed when the vessel proceeds to sea to avoid significant reduction in stability.
- 5) Tanks containing slack liquids should be kept to a minimum.
- 6) Compliance with the Stability Criteria incorporated in this booklet does not ensure immunity against capsizing, regardless of the circumstances, or absolve the Master from his responsibility.

The Master should therefore exercise prudence and good seamanship having regard to the season of the year, weather forecasts and the navigational zone, and should take the appropriate action as to speed and course warranted by the prevailing circumstances.

- 7) Before any stage of voyage commences, care should be taken to ensure that sizeable pieces of equipment have been properly stowed and lashed so as to minimise the possibility of both longitudinal and lateral shifting while at sea, under the effect of acceleration caused by rolling and pitching.

TANK CAPACITY TABLE

COMPARTMENT	LOCATION	CAPACITY (M ³)	S.G.	LCG FROM A.P. (M)	VCG FROM B (M)	FREE SURFACE MOMENT (M ⁴)
F.P. BALLAST TANK (C)	FR.36 - F.E	16.76	1.025	19.179	2.321	13.06
F.O. TANK NO.1 (P)	FR.21 - FR.29	10.47	0.850	12.564	1.749	0.55
F.O. TANK NO.1 (S)	FR.21 - FR.29	10.47	0.850	12.564	1.749	0.55
F.O. TANK NO.2 (P)	FR.4 - FR.12	13.27	0.850	3.569	1.555	6.50
F.O. TANK NO.2 (S)	FR.4 - FR.12	13.27	0.850	3.569	1.555	6.50
F.O. DAY TANK (P)	FR.10 - FR.12	1.13	0.850	5.501	2.018	0.14
F.O. DAY TANK (S)	FR.10 - FR.12	1.13	0.850	5.501	2.018	0.14
F.W. TANK (P)	FR.30 1/2 - FR.36	5.70	1.000	16.559	2.279	0.36
F.W. TANK (S)	FR.30 1/2 - FR.36	5.70	1.000	16.559	2.279	0.36
F.W. HEADER TK (P)	FR.14 - FR.16 (ABOVE M. DK.)	0.76	1.000	7.500	3.007	0.1
F.W. HEADER TK (S)	FR.14 - FR.16 (ABOVE M. DK.)	0.76	1.000	7.500	3.007	0.1

HYDROSTATIC PROPERTIES

21.00M TWIN SCREW TUG

SHELL THK. = 8 MM

TRIM = 0.000 METRES

WATER DENSITY = 1.02522 TONNES/CUBIC METRES

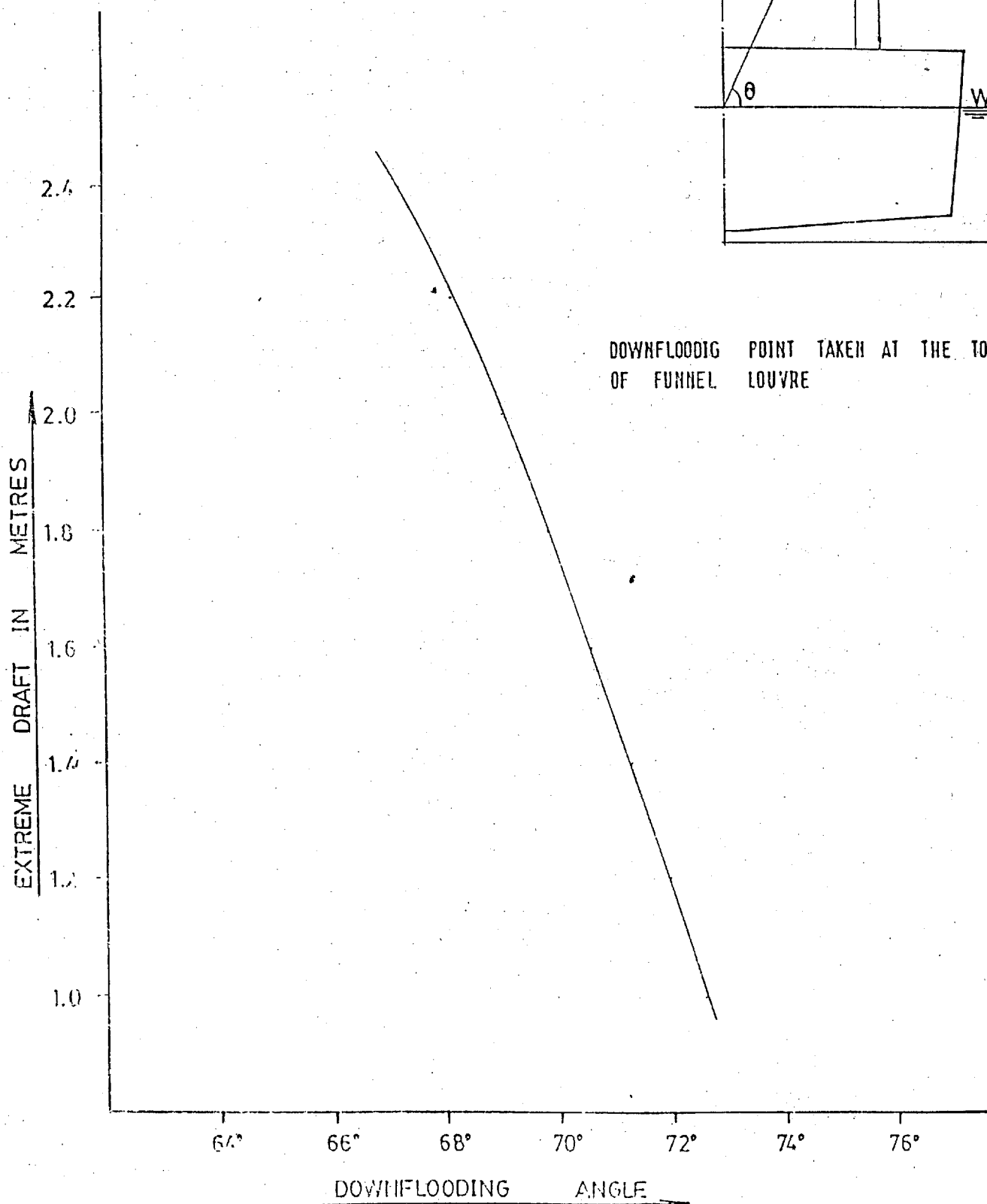
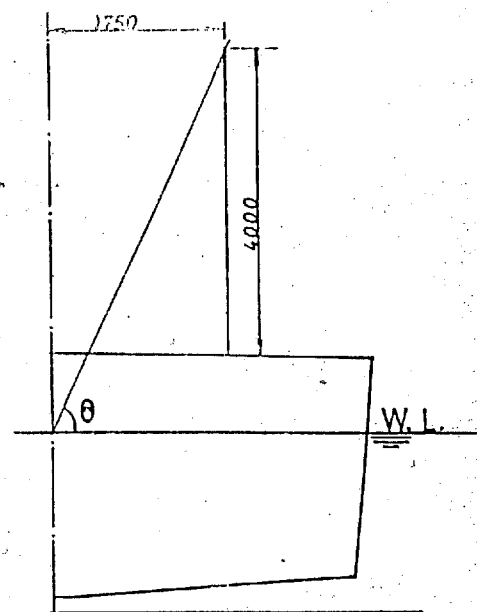
DRAFT TO U/S KEEL METRES	DISPL EXTREME TONNES	TONNES PER CM IMMERS-N TONNES	MOMENT TO CHANGE TRIM 1 CM TONNE METRES	VCB ABOVE BASE METRES	TRANSV MTC ABOVE BASE METRES	LONGIT-L CENTRE BUOYANCY FROM METRES	LONGIT-L CENTRE FLATAT-N A.P. METRES
0.500	18.03	0.765	0.51	0.360	13.468	9.37	9.91
0.600	25.70	0.834	0.61	0.419	11.055	9.58	10.09
0.700	34.11	0.870	0.71	0.475	8.605	9.71	10.20
0.800	43.08	0.886	0.79	0.530	6.362	9.80	10.24
0.900	52.45	0.894	0.86	0.585	4.568	9.86	10.24
1.000	62.05	0.908	0.94	0.640	3.467	9.90	10.22
1.100	71.70	0.939	1.01	0.696	3.301	9.94	10.20
1.200	81.25	1.001	1.09	0.754	4.314	10.01	10.19
1.300	91.71	1.030	1.19	0.810	4.248	10.00	10.05
1.400	102.52	1.063	1.31	0.867	4.180	9.98	9.87
1.500	113.67	1.104	1.45	0.925	4.020	9.93	9.63
1.600	125.14	1.155	1.63	0.983	3.677	9.88	9.33
1.700	136.73	1.170	1.68	1.040	3.563	9.83	9.29
1.800	148.48	1.169	1.67	1.097	3.421	9.80	9.39
1.900	160.29	1.174	1.69	1.153	3.326	9.76	9.43
2.000	172.10	1.184	1.72	1.207	3.260	9.74	9.45
2.100	184.02	1.192	1.74	1.262	3.206	9.73	9.49
2.200	196.02	1.200	1.76	1.317	3.164	9.71	9.52

21.00M TWIN SCREW TUG

ASSUMED KG = 0.00 M INITIAL TRIM = 0.00 M FREE TRIMMING ASSUMED
TABLE OF RIGHTING LEVERS (KN) - METRES WATER DENSITY = 1.02520 TONNES/CUB. METRE

DISPLACEMENT TONS	ANGLE OF HEEL - DEGREES							
	0.0	10.0	20.0	30.0	40.0	50.0	60.0	70.0
20.00	0.000	1.441	2.035	2.305	2.409	2.405	2.337	2.206
30.00	0.000	1.253	1.864	2.196	2.384	2.437	2.397	2.256
40.00	0.000	1.100	1.747	2.114	2.349	2.442	2.414	2.269
50.00	0.000	0.980	1.664	2.054	2.308	2.426	2.400	2.256
60.00	0.000	0.887	1.599	2.009	2.267	2.393	2.368	2.232
70.00	0.000	0.818	1.531	1.975	2.230	2.347	2.332	2.208
80.00	0.000	0.766	1.474	1.936	2.187	2.297	2.288	2.177
90.00	0.000	0.726	1.421	1.883	2.138	2.246	2.243	2.143
100.00	0.000	0.695	1.372	1.833	2.086	2.193	2.195	2.106
110.00	0.000	0.669	1.327	1.782	2.032	2.140	2.146	2.068
120.00	0.000	0.647	1.281	1.729	1.973	2.083	2.095	2.028
130.00	0.000	0.627	1.233	1.672	1.907	2.021	2.041	1.986
140.00	0.000	0.610	1.183	1.611	1.837	1.956	1.987	1.944
150.00	0.000	0.596	1.136	1.540	1.767	1.893	1.934	1.905
160.00	0.000	0.583	1.091	1.461	1.699	1.832	1.885	1.869
170.00	0.000	0.570	1.042	1.387	1.632	1.772	1.834	1.832
180.00	0.000	0.557	0.989	1.318	1.568	1.716	1.782	1.795
190.00	0.000	0.539	0.936	1.252	1.501	1.658	1.732	1.760
200.00	0.000	0.514	0.884	1.189	1.429	1.598	1.684	1.728

'KRESNA DUA'
CURVE OF DOWNFLOODING ANGLE

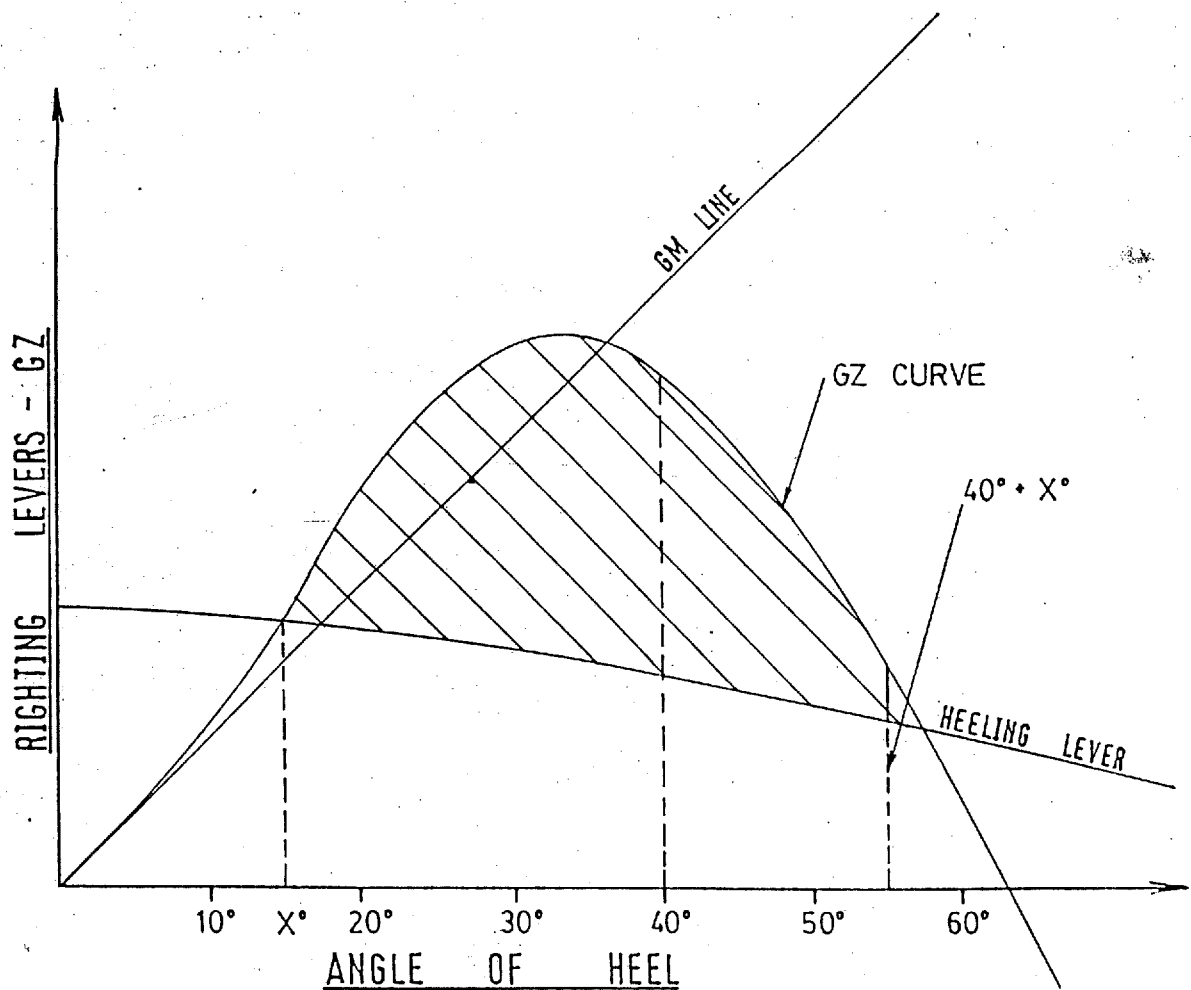


STABILITY CRITERIA

The Stability Criteria incorporated in this Stability Booklet are as follows:

- 1) The area under the GZ curve up to 30° to be not less than 0.055 Metre-Radians (10.34 Ft-Degs);
- 2) The area under the GZ curve up to 40° or the Downflooding Angle, if this is less than 40° , to be not less than 0.09 Metre-Radians (16.92 Ft-Degs);
- 3) The area under the GZ curve between 30° and 40° or the Downflooding Angle, if this is less than 40° , to be not less than 0.03 Metre-Radians (5.64 Ft-Degs);
- 4) The Righting Lever GZ should be at least 0.20 Metre (0.66 Ft at an Angle of Heel equal to or greater than 30° ;
- 5) The maximum righting arm should occur at an Angle of Heel preferably exceeding 30° but not less than 25° ;
- 6) The initial Metacentric height, GM, to be not less than 0.15 Metre (0.49 Ft).

CRITERIA USED FOR TOWING CONDITIONS



- (a) If downflooding angle is equal to or greater than $40^\circ + X^\circ$ (Where X° is first intercept between GZ and heeling lever curves), shaded area between GZ and heeling lever curves (i.e. the residual dynamic stability) to be not less than 0.090 m-radians.
- (b) If downflooding angle is less than $40^\circ + X^\circ$ shaded area (up to down flooding angle) to be not less than 0.090 m-radians.

CONDITION NO. 1
LIGHTSHIP

ITEM	WEIGHT	V.C.G.	VERT-L	L.C.G.	LONG-L	FREE
	ABOVE	MOMENT	FROM	MOMENT	SURF.	MOMENT
	MLD.	ABOVE	AP	ABOUT	THE AP	MOMENT
	BASE	BASE				
	TONNES	METRES	TONNE	METRES	TONNE	TONNE
			METRES		METRES	METRES

DEADWEIGHT	0.0	0.00	0.0	0.0	0.0	0.0
LIGHTSHIP	88.8	2.43	216.2	10.0	884.8	0.0
DISPLACEMENT	88.8	2.43	216.2	10.0	884.8	0.0

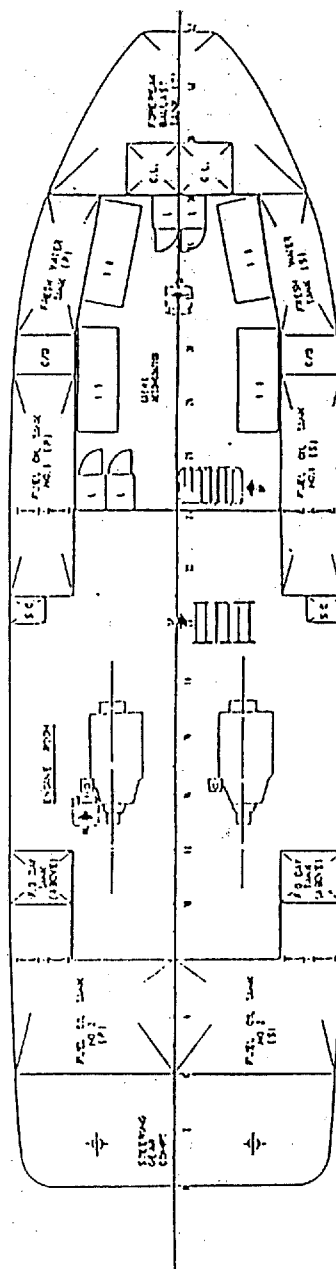
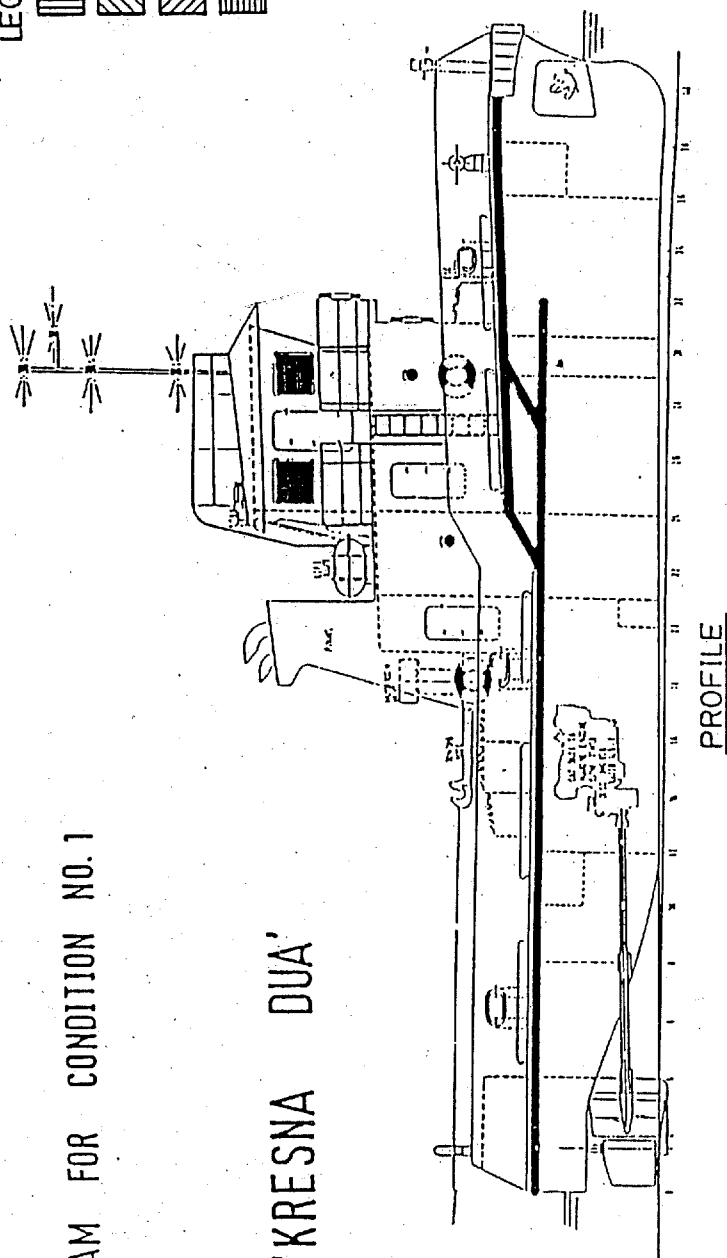
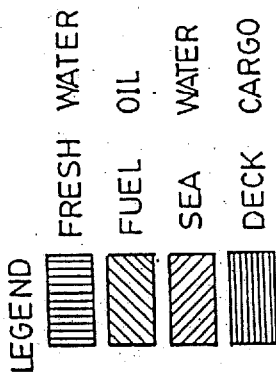
SAILING STATE

DENSITY OF WATER TAKEN AS 1.0252 (TONNES/CUBIC METRE)

FULL DRAFT AT L.C.F.	=	1.27 METRES
FULL DRAFT AFT AT MARKS	=	1.28 METRES
FULL DRAFT FWD AT MARKS	=	1.26 METRES
FULL MEAN DRAFT AT MIDSHIPS	=	1.27 METRES
TRIM BY STERN	=	0.03 METRES
K.M.T. ABOVE MLD. BASE	=	4.16 METRES
K.G. ABOVE MLD. BASE	=	2.43 METRES
G.M. (SOLID)	=	1.73 METRES
FREE SURFACE CORRECTION	=	0.00 METRES
G.M. (FLUID)	=	1.73 METRES

LOADING DIAGRAM FOR CONDITION NO. 1

'KRESNA DUA'



CONDITION NO. 1
LIGHTSHIP

STABILITY INFORMATION

K G ABOVE MLD. BASE	=	2.43 METRES
FREE SURFACE CORRECTION	=	0.00 METRES
KG CORRECTED(VIRTUAL KG)	=	2.43 METRES
MAXIMUM ALLOWABLE KG	=	2.32 METRES

ANGLE OF HEEL DEGREES	RIGHTING LEVER(KN) METRES	KG X SIN(ANGLE) METRES	RIGHTING LEVER(G) METRES
0.000	0.000	0.000	0.000
10.000	0.730	0.423	0.307
20.000	1.427	0.833	0.594
30.000	1.888	1.217	0.671
40.000	2.143	1.565	0.578
50.000	2.253	1.865	0.388
60.000	2.249	2.109	0.140

AREA UNDER CURVE UP TO 30.0 DEGREES	=	0.222 METRE RADIANS
FLOOD ANGLE NOT LESS THAN 40.0 DEGREES		
AREA UNDER CURVE UP TO 40.0 DEGREES	=	0.333 METRE RADIANS
AREA BETWEEN 30.0 AND 40.0 DEGREES	=	0.111 METRE RADIANS
MAXIMUM RIGHTING LEVER(GZ)	=	0.671 METRES
AND OCCURS AT 29.0 DEGREES		
MAXIMUM VERTICAL DEADWT MOMENT	=	-10.0
ACTUAL VERTICAL DEADWT MOMENT	=	0.0

LOADING CONDITION

STABILITY CURVE

Ship: P64

Yard: GREENBAY MARINE PTE LTD

Date: 2-11-91

CONDITION NO. 1

LIGHTSHIP

AREAS (m.rad) :

0.0° - 30.0° : 0.222

0.0° - 40.0° : 0.333

30.0° - 40.0° : 0.111

GZMAX

0.571

GZMAX AT

29.0°

GM

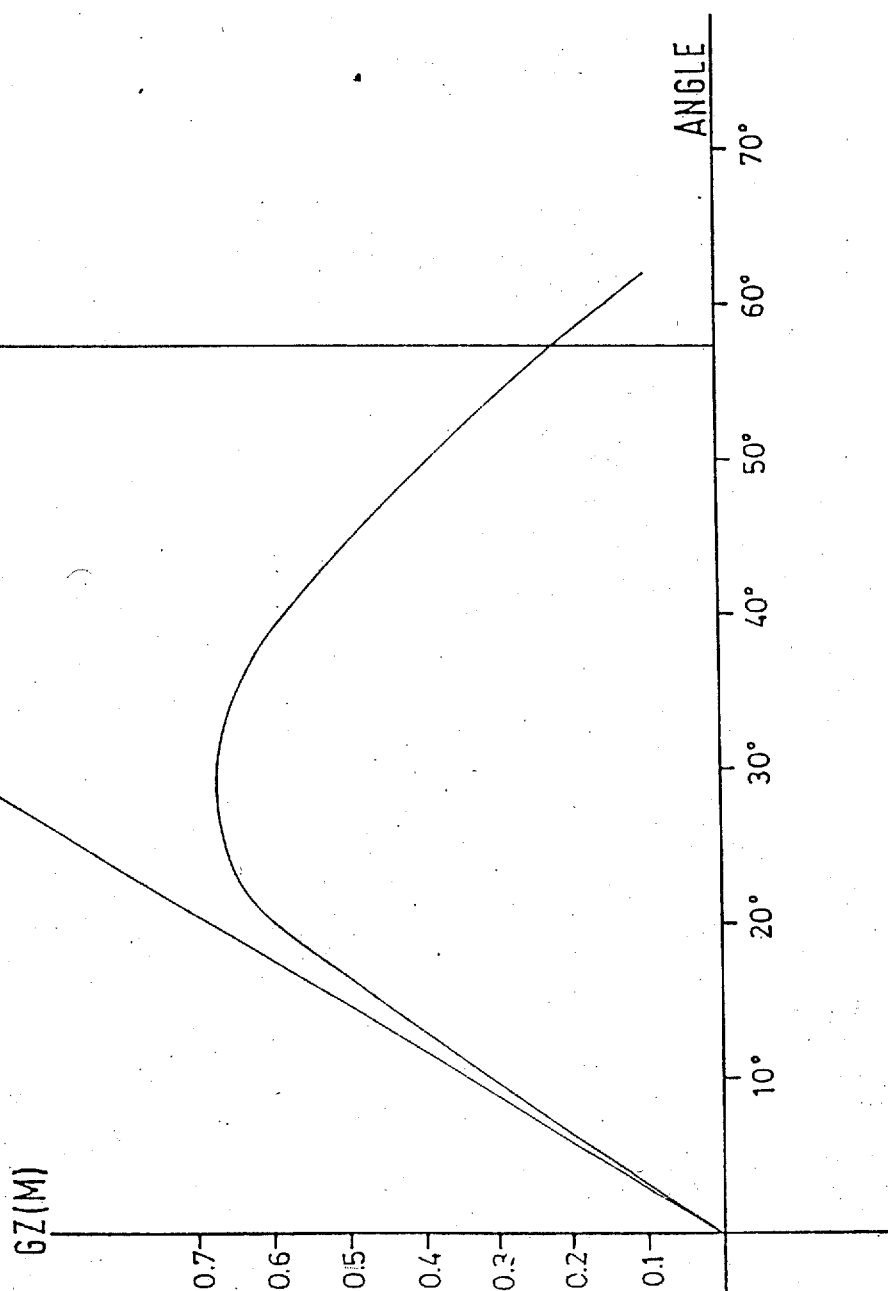
1.73

FLOOD ANGLE >=

40.0°

ACT V DWT MOM

3.0



CONDITION NO. 2
FULLY LOADED TO LOADLINE DRAFT @1.798M

ITEM	WEIGHT TONNES	V.C.G. ABOVE MLD. BASE METRES	VERT-L MOMENT ABOVE BASE TONNE METRES	L.C.G. FROM AP METRES	LONG-L MOMENT ABOUT THE AP TONNE METRES	FREE SURF. MOMENT TONNE METRES
CREW & EFFECTS	0.5	2.70	1.4	14.5	7.3	0.0
FUEL OIL DAY TK (P&S)	1.9	2.02	3.9	5.5	10.6	0.2
FUEL OIL TK NO.1 (P&S)	17.8	1.75	31.1	12.6	223.6	0.9
FUEL OIL TK NO.2 (P&S)	22.6	1.56	35.2	3.6	80.5	11.0
FRESH WATER TK (P&S)	11.4	2.28	26.0	16.6	188.8	0.7
F. W. HEADER TK (P&S)	1.5	3.01	4.6	7.5	11.4	0.1
FOREPEAK BALLAST TK (C)	3.7	1.11	4.2	18.8	70.5	13.4

DEADWEIGHT	59.4	1.79	106.3	10.0	592.6	26.3
LIGHTSHIP	88.8	2.43	216.2	10.0	884.8	0.0
DISPLACEMENT	148.2	2.18	322.5	10.0	1477.4	26.3

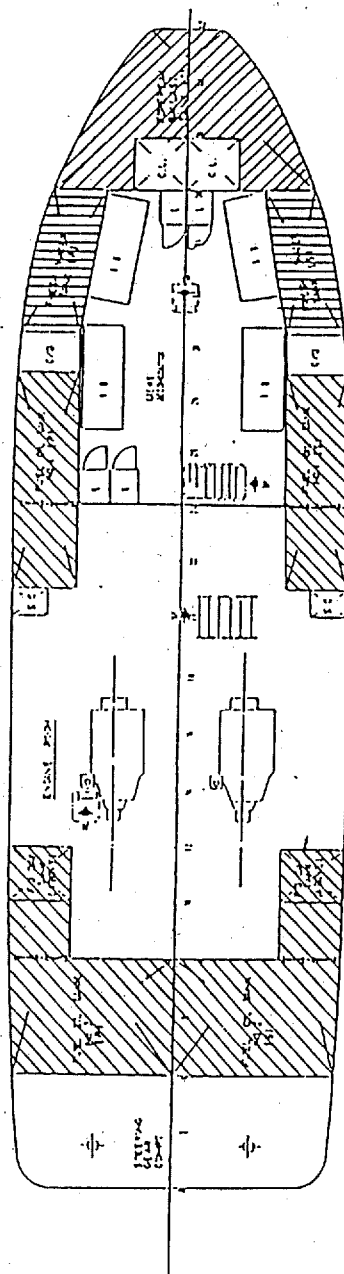
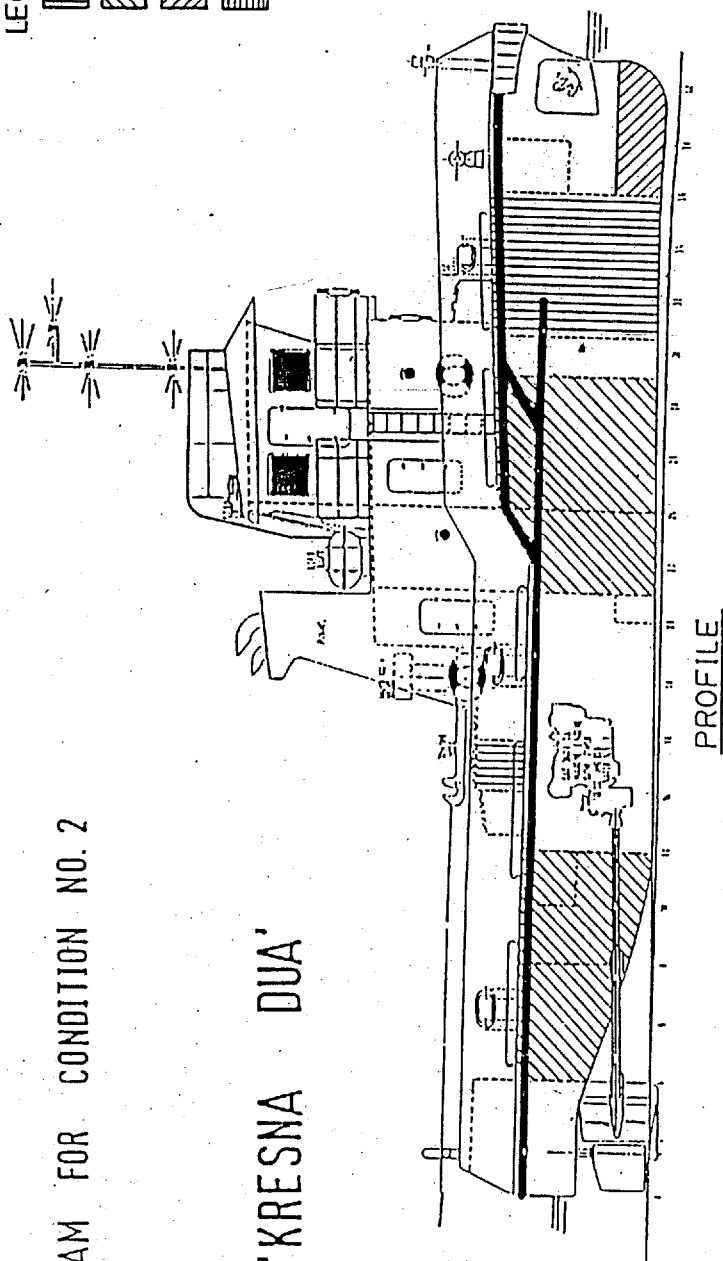
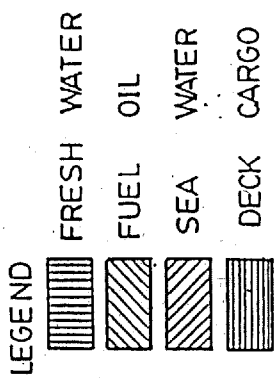
SAILING STATE

DENSITY OF WATER TAKEN AS 1.0252 (TONNES/CUBIC METRE)

FULL DRAFT AT L.C.F.	=	1.80 METRES
FULL DRAFT AFT AT MARKS	=	1.74 METRES
FULL DRAFT FWD AT MARKS	=	1.86 METRES
FULL MEAN DRAFT AT MIDSHIPS	=	1.80 METRES
TRIM BY BOW	=	0.15 METRES
K.M.T. ABOVE MLD. BASE	=	3.42 METRES
K.G. ABOVE MLD. BASE	=	2.18 METRES
G.M. (SOLID)	=	1.25 METRES
FREE SURFACE CORRECTION	=	0.18 METRES
G.M. (FLUID)	=	1.07 METRES

LOADING DIAGRAM FOR CONDITION NO. 2

'KRESNA DUA'



BELOW MAIN DECK

CONDITION NO. 2

FULLY LOADED TO LOADLINE DRAFT @1.798M

STABILITY INFORMATION FOR SAILING STATE

K G ABOVE MLD. BASE	=	2.18 METRES
FREE SURFACE CORRECTION	=	0.18 METRES
KG CORRECTED(VIRTUAL KG)	=	2.35 METRES
MAXIMUM ALLOWABLE KG	=	2.08 METRES

ANGLE OF HEEL DEGREES	RIGHTING LEVER(KN) METRES	KG X SIN(ANGLE) METRES	RIGHTING LEVER(GZ) METRES
0.000	0.000	0.000	0.000
10.000	0.598	0.409	0.190
20.000	1.146	0.805	0.341
30.000	1.549	1.177	0.372
40.000	1.780	1.513	0.267
50.000	1.905	1.803	0.102
60.000	1.945	2.038	-0.093

AREA UNDER CURVE UP TO 30.0 DEGREES	=	0.128 METRE RADIAN
FLOOD ANGLE NOT LESS THAN 40.0 DEGREES		
AREA UNDER CURVE UP TO 40.0 DEGREES	=	0.186 METRE RADIAN
AREA BETWEEN 30.0 AND 40.0 DEGREES	=	0.057 METRE RADIAN
MAXIMUM RIGHTING LEVER(GZ)	=	0.376 METRES
AND OCCURS AT 27.0 DEGREES		
MAXIMUM VERTICAL DEADWT MOMENT	=	91.5
ACTUAL VERTICAL DEADWT MOMENT	=	132.6

CONDITION NO. 2

FULLY LOADED TO LOADLINE DRAFT @1.798M

STABILITY INFORMATION FOR TOWING STATE

K B ABOVE MLD. BASE = 1.096 METRES

HEIGHT OF TOW HOOK ABOVE BASELINE = 3.865 METRES

HEELING MOMENT = $T \times AD \times \cos(\text{ANGLE})$ WHERE T = ASSUMED 1/2 OF MAX. BOLLARD PULL
= 5.0 TONNESAD = VERTICAL HEIGHT OF TOWING POSITION
FROM KB
= 2.769 METRES

HEELING LEVER = HEELING MOMENT / DISPLACEMENT OF VESSEL

ANGLE OF HEEL DEGREES	AD x COS(ANGLE) METRES	HEELING MOMENT TONNE-METRES	HEELING LEVER METRES
0.000	2.769	13.845	0.093
10.000	2.727	13.635	0.092
20.000	2.602	13.010	0.088
30.000	2.398	11.990	0.081
40.000	2.121	10.606	0.072
50.000	1.780	8.899	0.060
60.000	1.385	6.923	0.047

THE AREA OF THE RESIDUAL DYNAMIC
STABILITY UP TO ANGLE OF HEEL OF
40.0 DEGREES PLUS THE ANGLE OF
FIRST INTERCEPT
(DOWNFLOODING ANGLE > 40 DEGREES)

= 0.143 METRE RADIANS

LOADING CONDITION

STABILITY CURVE

Ship: P64

Yard: GREENBAY MARINE PTE LTD

Date: 2-11-91

CONDITION NO. 2

FULLY LOADED TO LOADLINE; DRAFT
@ 1.798M

AREAS (m.rad) :

0.0° - 30.0° : 0.128

0.0° - 40.0° : 0.186

30.0° - 40.0° : 0.057

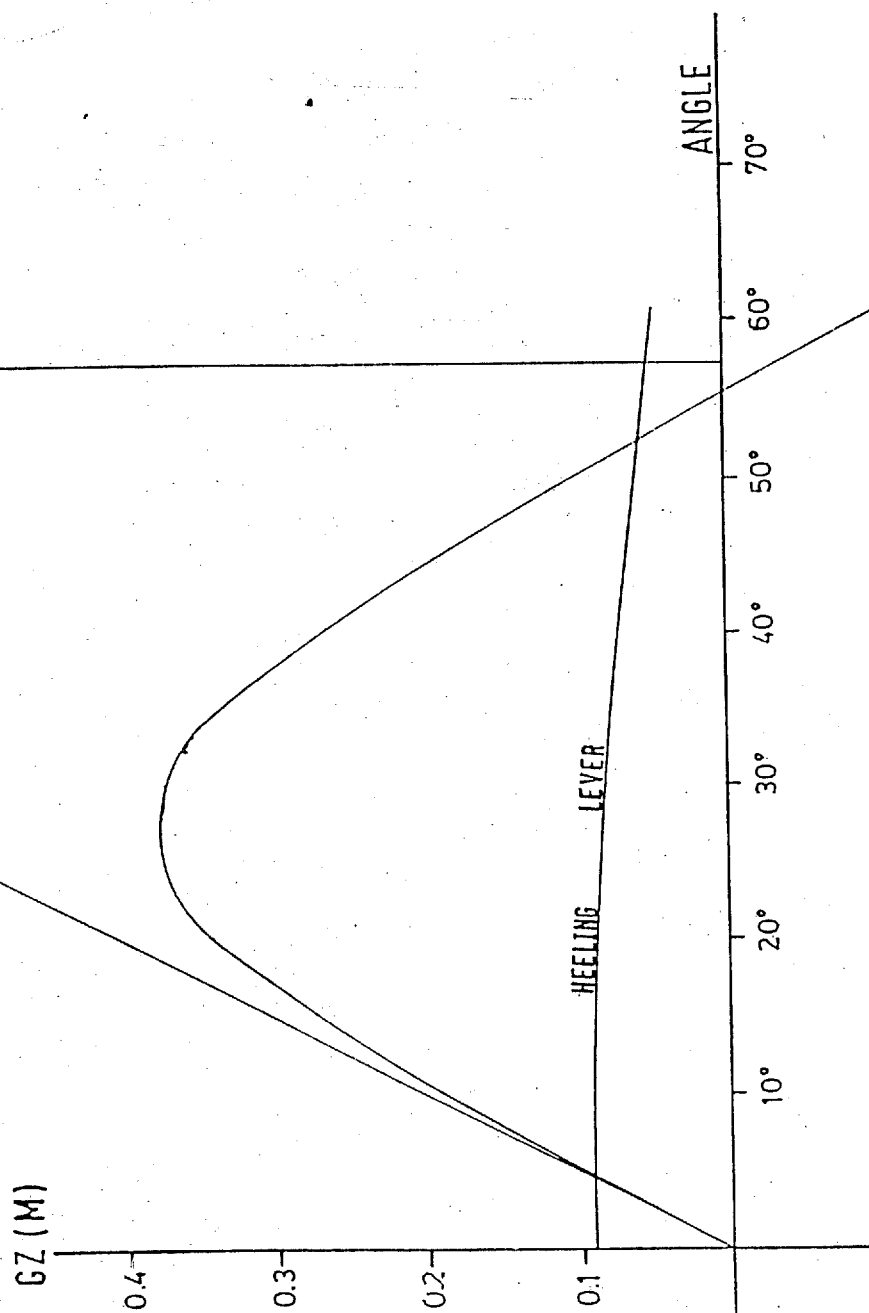
GZMAX 0.376 m

GZMAX AT 27.0°

GM 1.07

FLOOD ANGLE >= 40.0°

ACT V DWT MOM 132.0



CONDITION NO. 3

FULLY LOADED ARRIVAL FROM CONDITION NO.2

ITEM	WEIGHT TONNES	V.C.G. ABOVE MLD. BASE METRES	VERT-L MOMENT ABOVE BASE TONNE METRES	L.C.G. FROM AP METRES	LONG-L MOMENT ABOUT THE AP TONNE METRES	FREE SURF. MOMENT TONNE METRES
CREW & EFFECTS	0.5	2.70	1.4	14.5	7.3	0.0
FUEL OIL DAY TK (P&S)	1.9	2.02	3.9	5.5	10.6	0.2
FUEL OIL TK NO.1 (P&S)	1.8	0.70	1.2	12.3	21.8	0.9
FUEL OIL TK NO.2 (P&S)	2.3	0.72	1.6	4.4	9.9	11.0
FRESH WATER TK (P&S)	1.1	1.25	1.4	16.4	18.6	0.7
F. W. HEADER TK (P&S)	1.5	3.01	4.6	7.5	11.4	0.1
FOREPEAK BALLAST TK (C)	3.7	1.11	4.2	18.8	70.5	13.4

DEADWEIGHT	12.9	1.42	18.3	11.7	150.0	26.3
LIGHTSHIP	88.8	2.43	216.2	10.0	884.8	0.0
DISPLACEMENT	101.7	2.31	234.5	10.2	1034.8	26.3

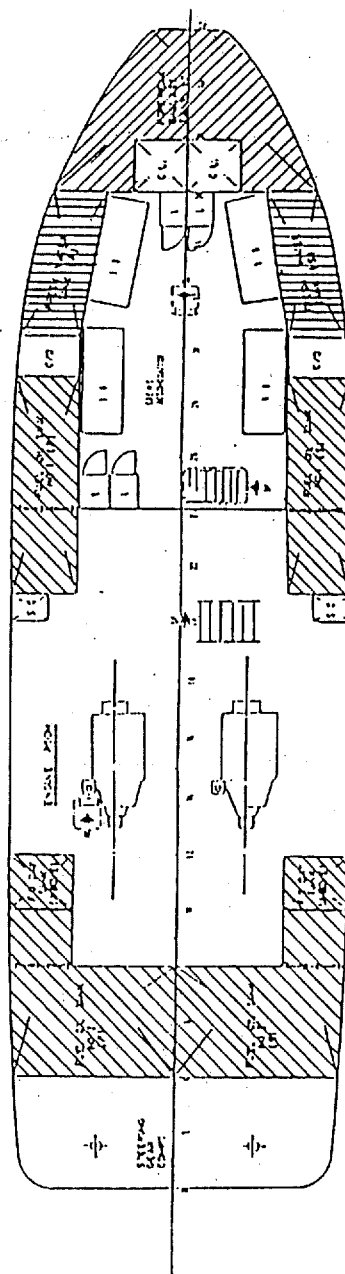
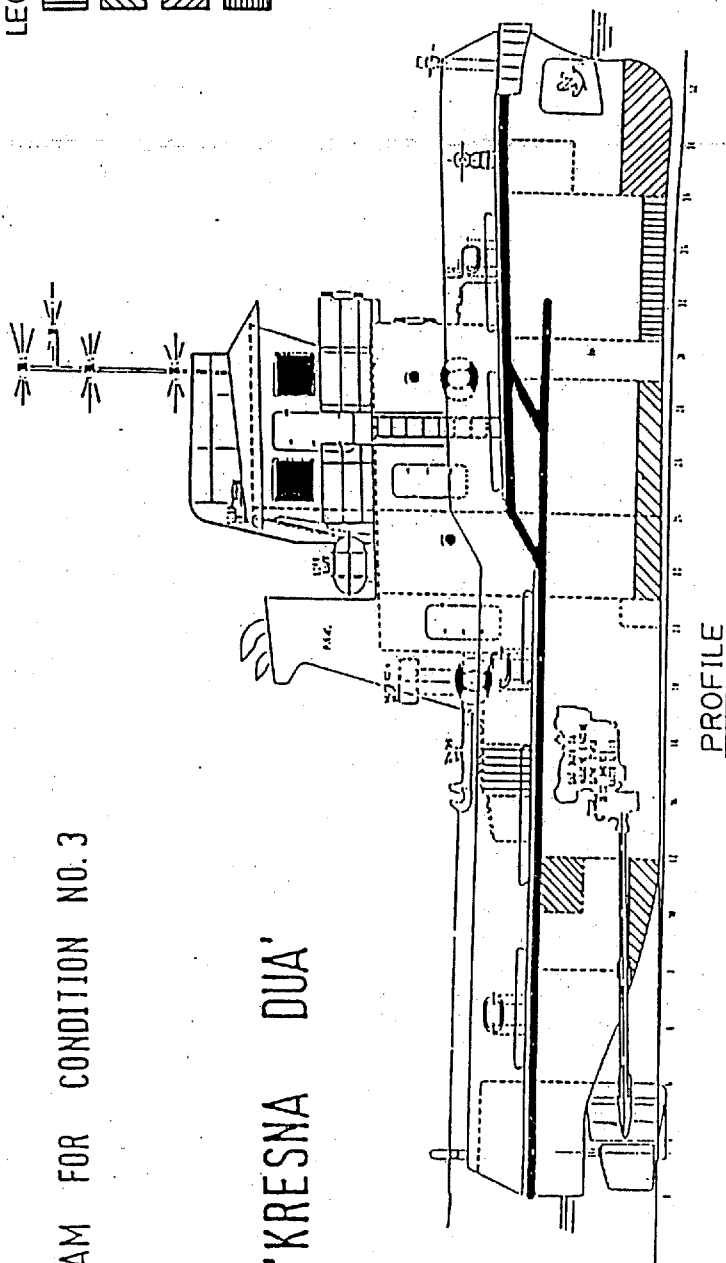
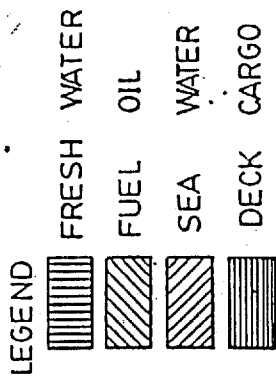
SAILING STATE

DENSITY OF WATER TAKEN AS 1.0252 (TONNES/CUBIC METRE)

FULL DRAFT AT L.C.F.	=	1.39 METRES
FULL DRAFT AFT AT MARKS	=	1.32 METRES
FULL DRAFT FWD AT MARKS	=	1.47 METRES
FULL MEAN DRAFT AT MIDSHIPS	=	1.40 METRES
TRIM BY BOW	=	0.16 METRES
K.M.T. ABOVE MLD. BASE	=	3.96 METRES
K.G. ABOVE MLD. BASE	=	2.31 METRES
G.M. (SOLID)	=	1.65 METRES
FREE SURFACE CORRECTION	=	0.26 METRES
G.M. (FLUID)	=	1.40 METRES

LOADING DIAGRAM FOR CONDITION NO.3

'KRESNA DUA'



BELOW MAIN DECK

CONDITION NO. 3
FULLY LOADED ARRIVAL FROM CONDITION NO.2

STABILITY INFORMATION FOR SAILING STATE

K G ABOVE MLD. BASE = 2.31 METRES
FREE SURFACE CORRECTION = 0.26 METRES
KG CORRECTED(VIRTUAL KG) = 2.57 METRES
MAXIMUM ALLOWABLE KG = 2.32 METRES

ANGLE OF HEEL DEGREES	RIGHTING LEVER(KN) METRES	KG X SIN(ANGLE) METRES	RIGHTING LEVER(GZ) METRES
0.000	0.000	0.000	0.000
10.000	0.689	0.445	0.244
20.000	1.365	0.877	0.488
30.000	1.825	1.283	0.543
40.000	2.079	1.649	0.430
50.000	2.186	1.965	0.221
60.000	2.188	2.222	-0.034

AREA UNDER CURVE UP TO 30.0 DEGREES = 0.179 METRE RADIAN
FLOOD ANGLE NOT LESS THAN 40.0 DEGREES
AREA UNDER CURVE UP TO 40.0 DEGREES = 0.266 METRE RADIAN
AREA BETWEEN 30.0 AND 40.0 DEGREES = 0.087 METRE RADIAN
MAXIMUM RIGHTING LEVER(GZ) = 0.546 METRES
AND OCCURS AT 28.0 DEGREES
MAXIMUM VERTICAL DEADWT MOMENT = 19.8
ACTUAL VERTICAL DEADWT MOMENT = 44.6

CONDITION NO. 3
FULLY LOADED ARRIVAL FROM CONDITION NO.2

STABILITY INFORMATION FOR TOWING STATE

K B ABOVE MLD. BASE = 0.861 METRES
HEIGHT OF TOW HOOK ABOVE BASELINE = 3.865 METRES

HEELING MOMENT = $T \times AD \times \cos(\text{ANGLE})$

WHERE T = ASSUMED 1/2 OF MAX. BOLLARD PULL
= 5.0 TONNES

AD = VERTICAL HEIGHT OF TOWING POSITION
FROM KB
= 3.004 METRES

HEELING LEVER = HEELING MOMENT / DISPLACEMENT OF VESS

ANGLE OF HEEL DEGREES	$AD \times \cos(\text{ANGLE})$ METRES	HEELING MOMENT TONNE-METRES	HEELING LEV METRES
0.000	3.004	15.020	0.148
10.000	2.958	14.792	0.145
20.000	2.823	14.114	0.139
30.000	2.602	13.008	0.128
40.000	2.301	11.506	0.113
50.000	1.931	9.655	0.095
60.000	1.502	7.510	0.074

THE AREA OF THE RESIDUAL DYNAMIC
STABILITY UP TO ANGLE OF HEEL OF
40.0 DEGREES PLUS THE ANGLE OF
FIRST INTERCEPT = 0.208 METRE RADIAN
(DOWNFLOODING ANGLE > 40 DEGREES)

LOADING CONDITION

STABILITY CURVE

Ship: P64

Yard: GREENBAY MARINE PTE LTD.

Date: 2-11-91

CONDITION NO. 3
FULLY LOADED ARRIVAL FROM
CONDITION NO. 2

AREAS (m.rad) :

0.0° - 30.0° : 0.179

0.0° - 40.0° : 0.266

30.0° - 40.0° : 0.087

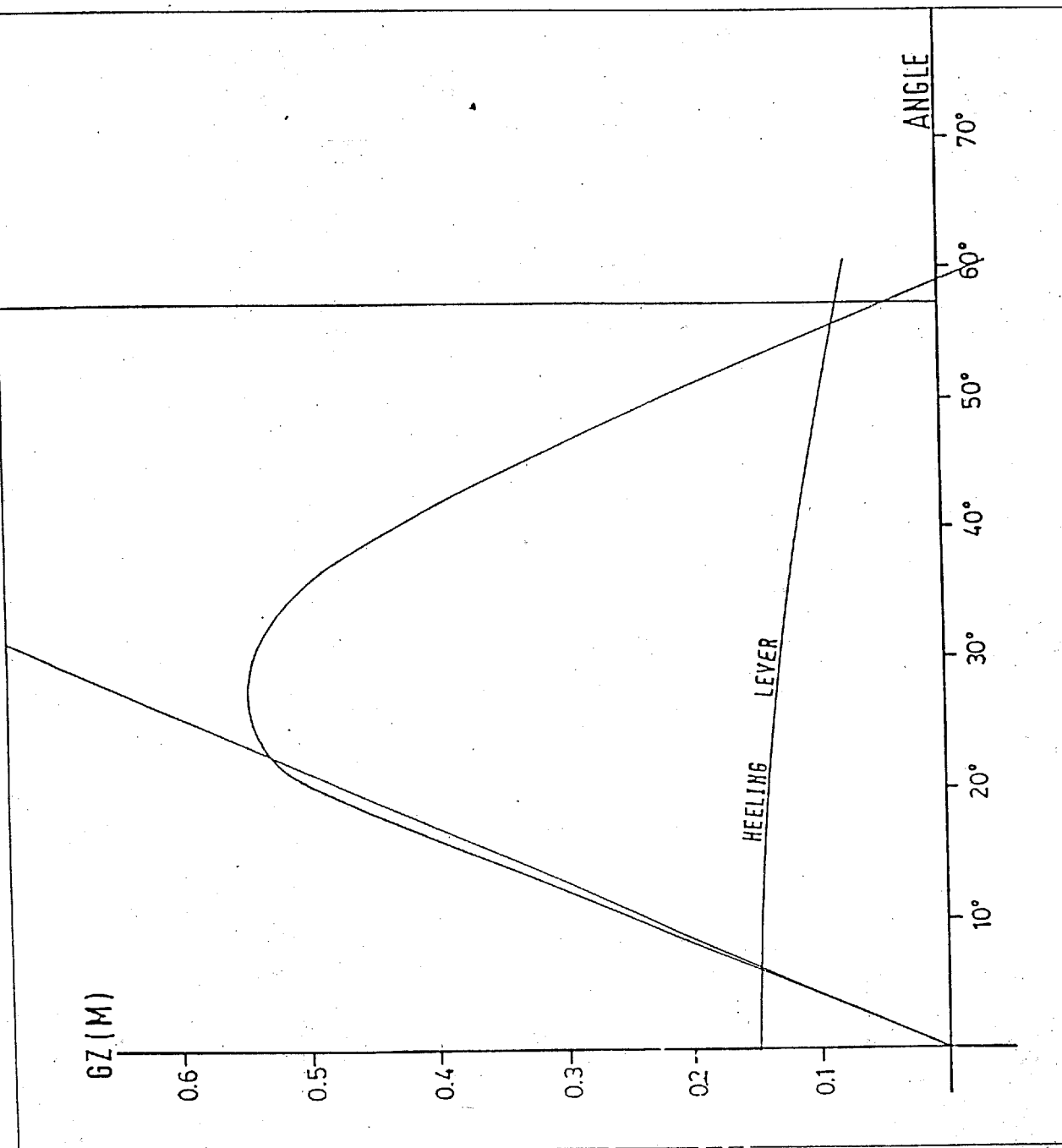
GZMAX 0.546 m

GZMAX AT 28.0°

GM 1.40

FLOOD ANGLE >= 40.0°

ACT V DHT MOM 44.0



CONDITION NO. 4

FULLY LOADED TO LOADLINE DRAFT @1.798M WITH DECK CARGO

ITEM	WEIGHT TONNES	V.C.G. ABOVE MLD. BASE METRES	VERT-L MOMENT ABOVE BASE TONNE METRES	L.C.G. FROM AP METRES	LONG-L MOMENT ABOUT THE AP TONNE METRES	FREE SURF. MOMENT TONNE METRES
CREW & EFFECTS	0.5	2.70	1.4	14.5	7.3	0.0
FUEL OIL DAY TK (P&S)	1.9	2.02	3.9	5.5	10.6	0.2
FUEL OIL TK NO.1 (P&S)	17.8	1.75	31.1	12.6	223.6	0.9
FUEL OIL TK NO.2 (P&S)	22.6	1.56	35.2	3.6	80.5	11.0
FRESH WATER TK (P&S)	11.4	2.28	26.0	16.6	188.8	0.7
F. W. HEADER TK (P&S)	1.5	3.01	4.6	7.5	11.4	0.1
DECK CARGO	3.7	3.13	11.7	4.0	15.0	0.0

DEADWEIGHT	59.4	1.92	113.8	9.0	537.1	12.9
LIGHTSHIP	88.8	2.43	216.2	10.0	884.8	0.0
DISPLACEMENT	148.2	2.23	330.1	9.6	1421.9	12.9

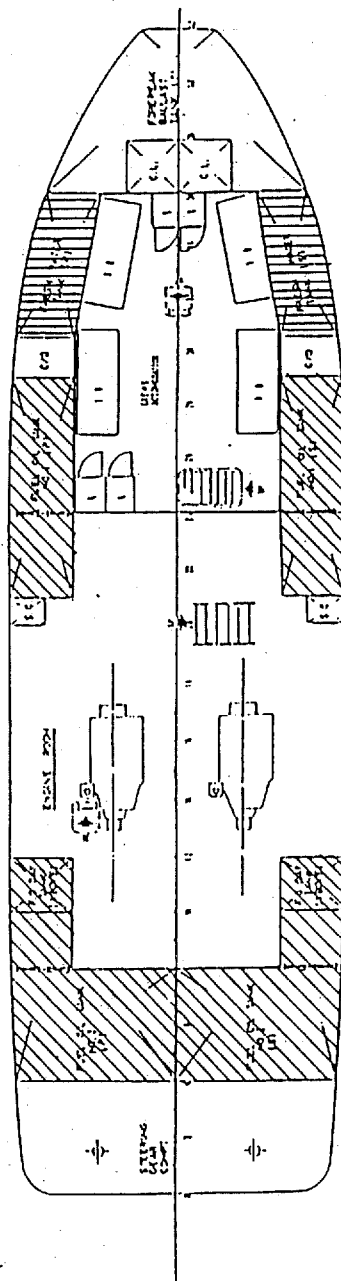
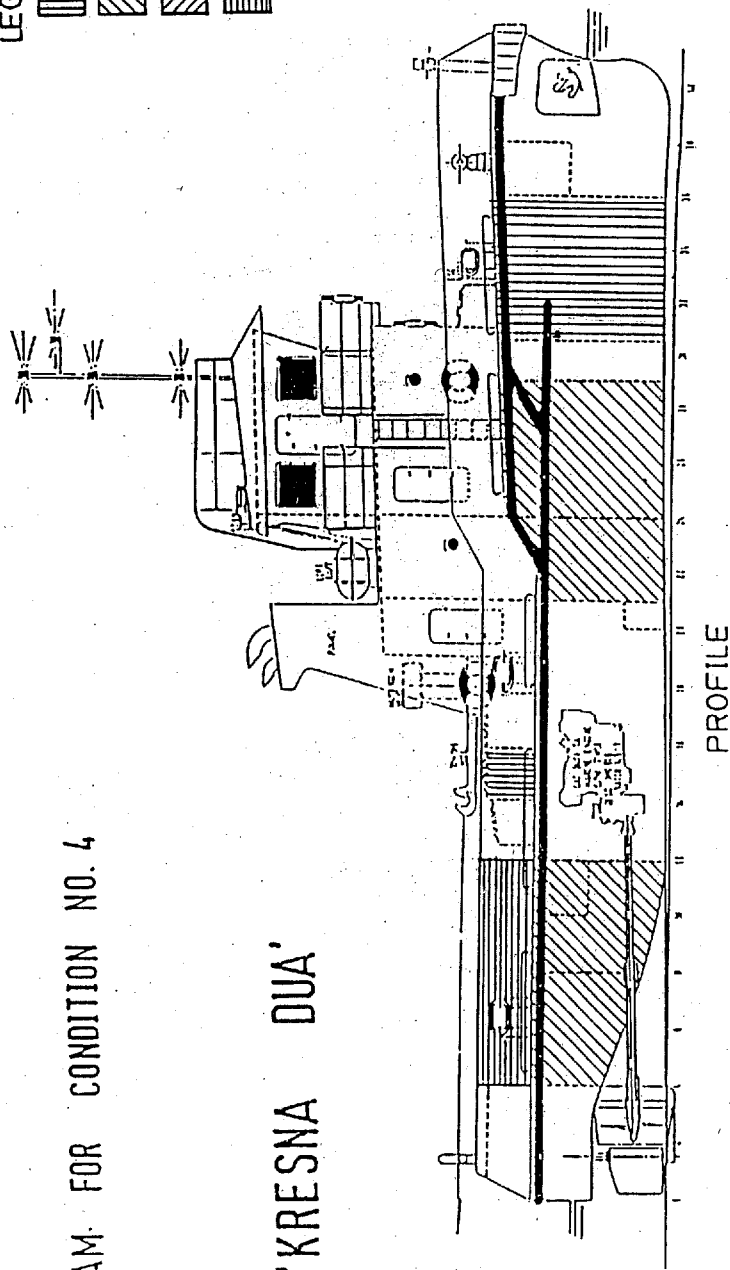
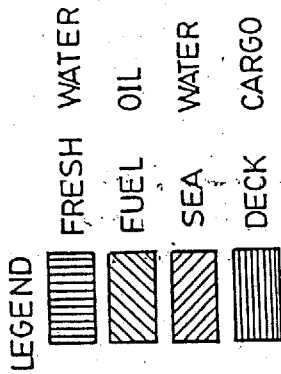
SAILING STATE

DENSITY OF WATER TAKEN AS 1.0252 (TONNES/CUBIC METRE)

FULL DRAFT AT L.C.F.	=	1.80 METRES
FULL DRAFT AFT AT MARKS	=	1.86 METRES
FULL DRAFT FWD AT MARKS	=	1.72 METRES
FULL MEAN DRAFT AT MIDSHIPS	=	1.79 METRES
TRIM BY STERN	=	0.18 METRES
K.M.T. ABOVE MLD. BASE	=	3.42 METRES
K.G. ABOVE MLD. BASE	=	2.23 METRES
G.M. (SOLID)	=	1.20 METRES
FREE SURFACE CORRECTION	=	0.09 METRES
G.M. (FLUID)	=	1.11 METRES

LOADING DIAGRAM FOR CONDITION NO. 4

'KRESNA DUA'



CONDITION NO. 4

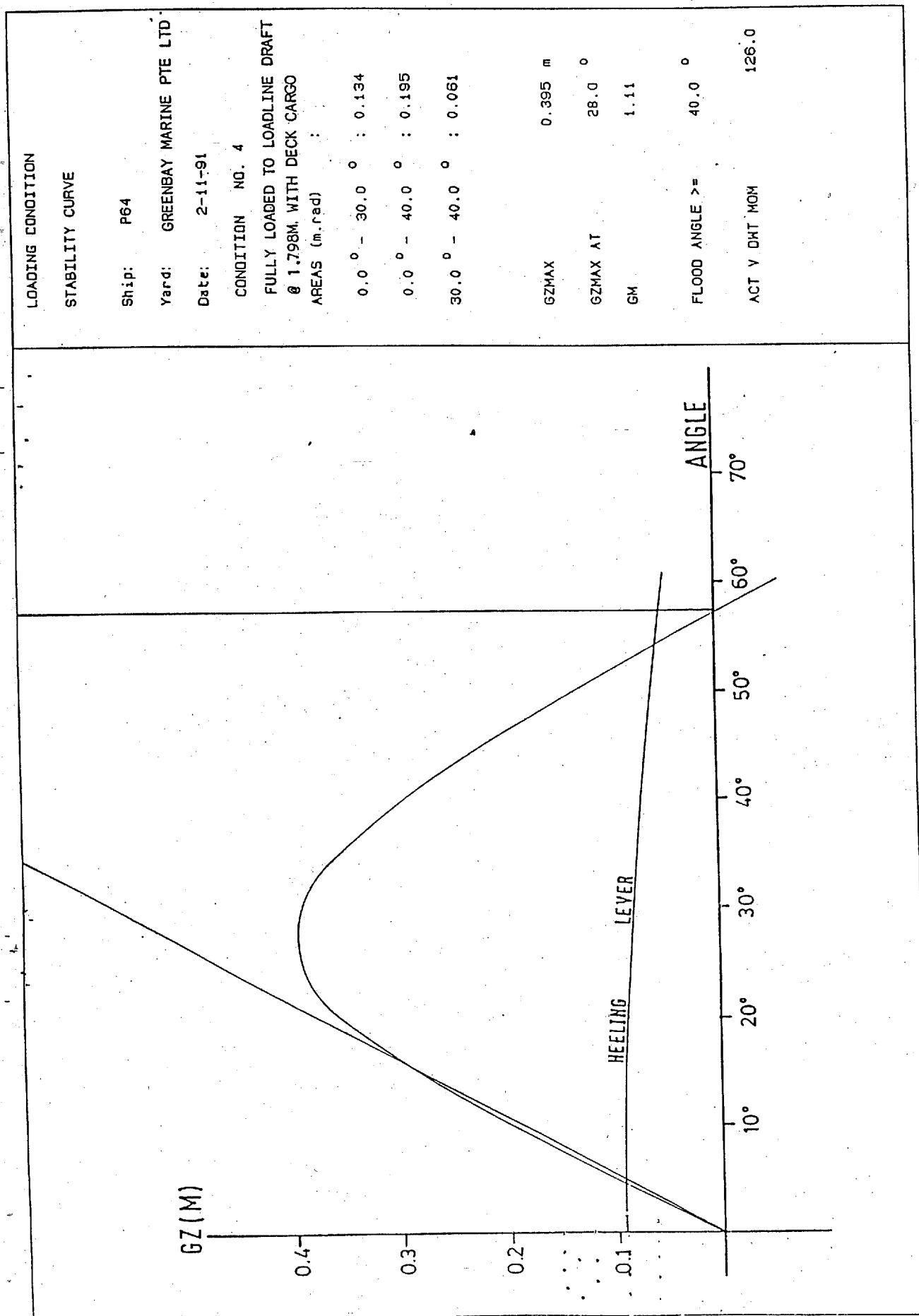
FULLY LOADED TO LOADLINE DRAFT @1.798M WITH DECK CARGO

STABILITY INFORMATION FOR SAILING STATE

K G ABOVE MLD. BASE	=	2.23 METRES
FREE SURFACE CORRECTION	=	0.09 METRES
KG CORRECTED(VIRTUAL KG)	=	2.31 METRES
MAXIMUM ALLOWABLE KG	=	2.08 METRES

ANGLE OF HEEL DEGREES	RIGHTING LEVER(KN) METRES	KG X SIN(ANGLE) METRES	RIGHTING LEVER(GZ) METRES
0.000	0.000	0.000	0.000
10.000	0.598	0.402	0.196
20.000	1.146	0.791	0.355
30.000	1.549	1.157	0.392
40.000	1.780	1.487	0.293
50.000	1.905	1.772	0.132
60.000	1.945	2.004	-0.059

AREA UNDER CURVE UP TO 30.0 DEGREES	=	0.134 METRE RADIANS
FLOOD ANGLE NOT LESS THAN 40.0 DEGREES		
AREA UNDER CURVE UP TO 40.0 DEGREES	=	0.195 METRE RADIANS
AREA BETWEEN 30.0 AND 40.0 DEGREES	=	0.061 METRE RADIANS
MAXIMUM RIGHTING LEVER(GZ)	=	0.395 METRES
AND OCCURS AT 28.0 DEGREES		
MAXIMUM VERTICAL DEADWT MOMENT	=	91.5
ACTUAL VERTICAL DEADWT MOMENT	=	126.7



CONDITION NO. 5
FULLY LOADED ARRIVAL FROM CONDITION NO.4

ITEM	WEIGHT TONNES	V.C.G. ABOVE MLD. BASE METRES	VERT-L MOMENT ABOVE BASE TONNE METRES	L.C.G. FROM AP METRES	LONG-L MOMENT ABOUT THE AP TONNE METRES	FREE SURF. MOMENT TONNE METRES
CREW & EFFECTS	0.5	2.70	1.4	14.5	7.3	0.0
FUEL OIL DAY TK (P&S)	1.9	2.02	3.9	5.5	10.6	0.2
FUEL OIL TK NO.1 (P&S)	1.8	0.70	1.2	12.3	21.8	0.9
FUEL OIL TK NO.2 (P&S)	2.3	0.72	1.6	4.4	9.9	11.0
FRESH WATER TK (P&S)	1.1	1.25	1.4	16.4	18.6	0.7
F. W. HEADER TK (P&S)	1.5	3.01	4.6	7.5	11.4	0.1
DECK CARGO	3.7	3.13	11.7	4.0	15.0	0.0

DEADWEIGHT	12.9	2.01	25.8	7.3	94.5	12.9
LIGHTSHIP	88.8	2.43	216.2	10.0	884.8	0.0
DISPLACEMENT	101.7	2.38	242.0	9.6	979.3	12.9

SAILING STATE

DENSITY OF WATER TAKEN AS 1.0252 (TONNES/CUBIC METRE)

FULL DRAFT AT L.C.F.	=	1.39 METRES
FULL DRAFT AFT AT MARKS	=	1.51 METRES
FULL DRAFT FWD AT MARKS	=	1.26 METRES
FULL MEAN DRAFT AT MIDSHIPS	=	1.38 METRES
TRIM BY STERN	=	0.25 METRES
K.M.T. ABOVE MLD. BASE	=	3.96 METRES
K.G. ABOVE MLD. BASE	=	2.38 METRES
G.M. (SOLID)	=	1.58 METRES
FREE SURFACE CORRECTION	=	0.13 METRES
G.M. (FLUID)	=	1.45 METRES

CONDITION NO. 4

FULLY LOADED TO LOADLINE DRAFT @1.798M WITH DECK CARGO

STABILITY INFORMATION FOR TOWING STATE

K B ABOVE MLD. BASE = 1.096 METRES

HEIGHT OF TOW HOOK ABOVE BASELINE = 3.865 METRES

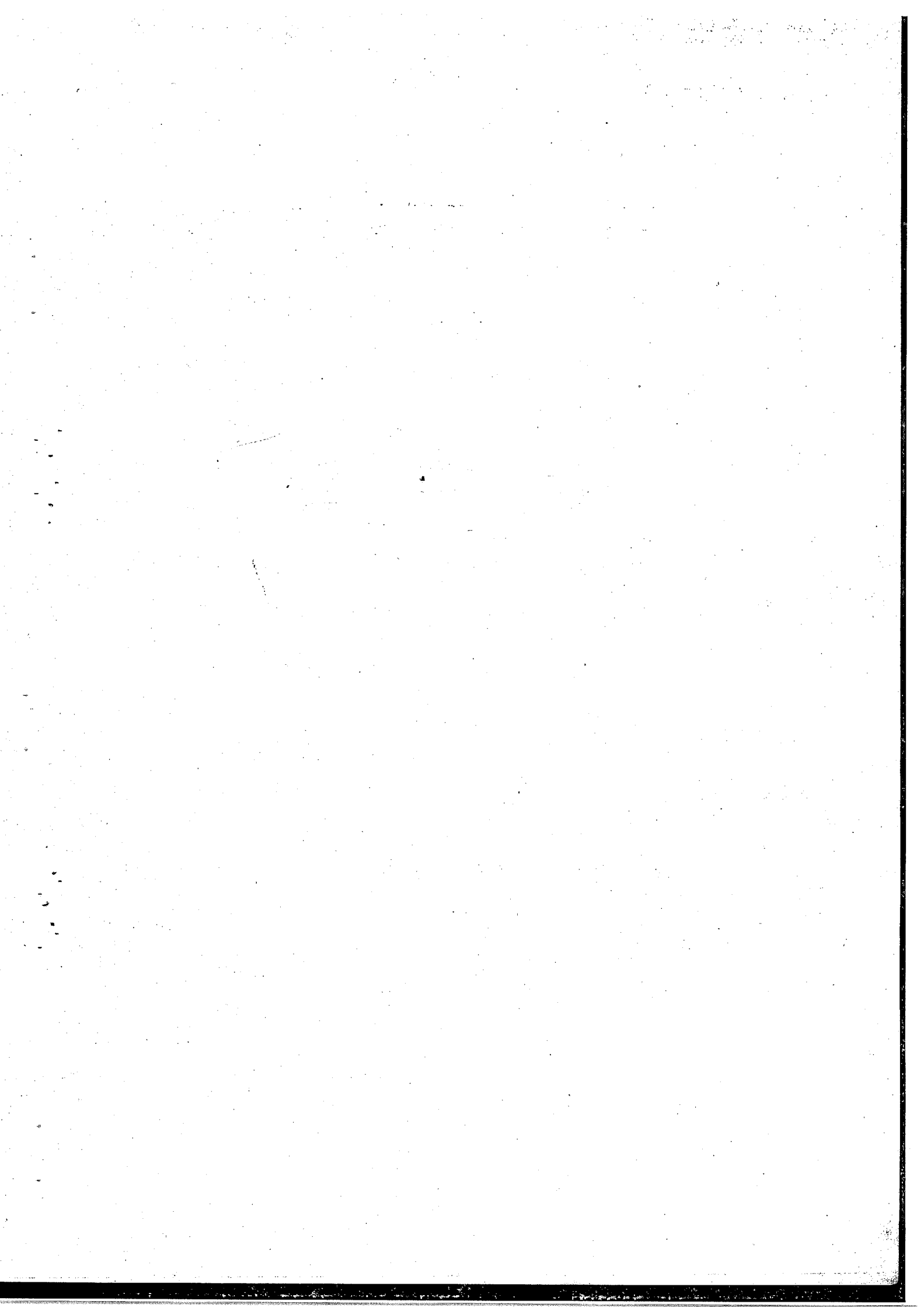
HEELING MOMENT = $T \times AD \times \cos(\text{ANGLE})$ WHERE T = ASSUMED 1/2 OF MAX. BOLLARD PULL
= 5.0 TONNES AD = VERTICAL HEIGHT OF TOWING POSITION
FROM KB
= 2.769 METRES

HEELING LEVER = HEELING MOMENT / DISPLACEMENT OF VESSEL

ANGLE OF HEEL DEGREES	$AD \times \cos(\text{ANGLE})$ METRES	HEELING MOMENT TONNE-METRES	HEELING LEVER METRES
0.000	2.769	13.845	0.093
10.000	2.727	13.635	0.092
20.000	2.602	13.010	0.088
30.000	2.398	11.990	0.081
40.000	2.121	10.606	0.072
50.000	1.780	8.899	0.060
60.000	1.385	6.923	0.047

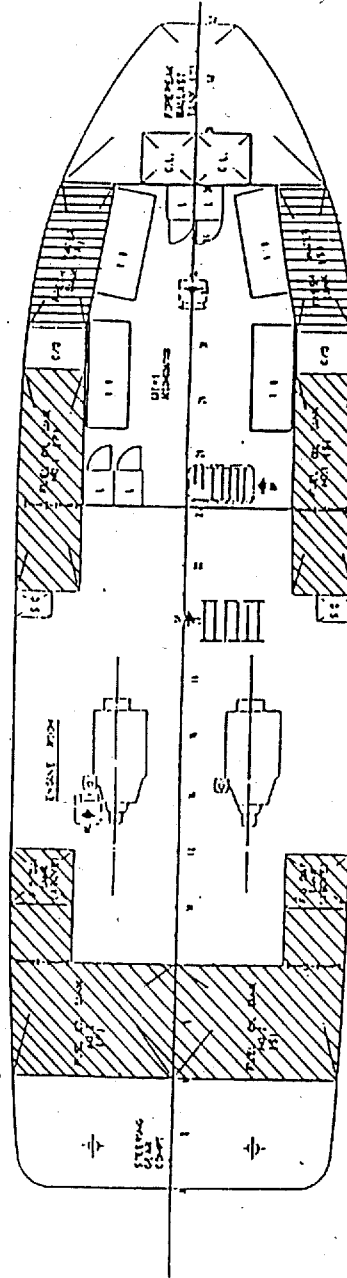
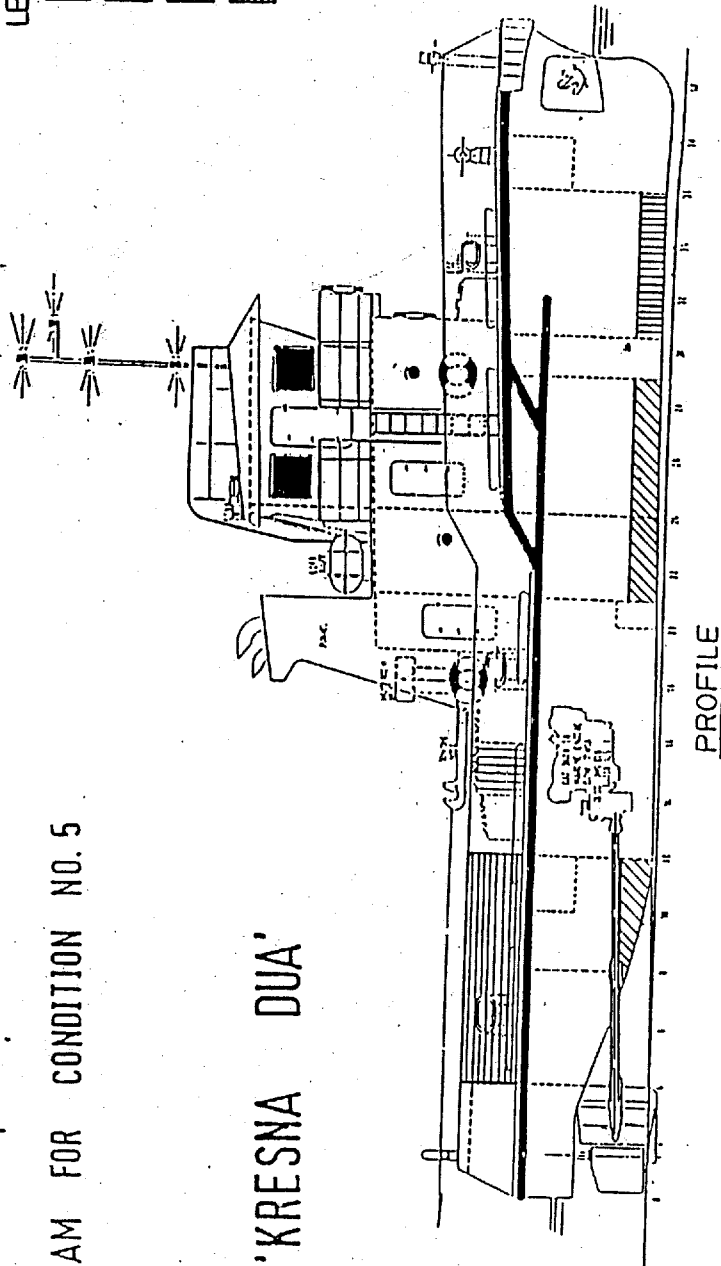
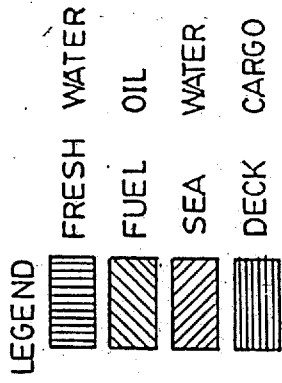
THE AREA OF THE RESIDUAL DYNAMIC
STABILITY UP TO ANGLE OF HEEL OF
40.0 DEGREES PLUS THE ANGLE OF
FIRST INTERCEPT
(DOWNFLOODING ANGLE > 40 DEGREES)

= 0.154 METRE RADIANS



LOADING DIAGRAM FOR CONDITION NO. 5

'KRESNA DUA'



CONDITION NO. 5
FULLY LOADED ARRIVAL FROM CONDITION NO.4

STABILITY INFORMATION FOR SAILING STATE

K G ABOVE MLD. BASE	=	2.38 METRES
FREE SURFACE CORRECTION	=	0.13 METRES
KG CORRECTED(VIRTUAL KG)	=	2.51 METRES
MAXIMUM ALLOWABLE KG	=	2.32 METRES

ANGLE OF HEEL DEGREES	RIGHTING LEVER(KN) METRES	KG X SIN(ANGLE) METRES	RIGHTING LEVER(GZ) METRES
0.000	0.000	0.000	0.000
10.000	0.689	0.435	0.253
20.000	1.365	0.858	0.507
30.000	1.825	1.254	0.571
40.000	2.079	1.612	0.467
50.000	2.186	1.921	0.265
60.000	2.188	2.172	0.016

AREA UNDER CURVE UP TO 30.0 DEGREES	=	0.187 METRE RADIANS
FLOOD ANGLE NOT LESS THAN 40.0 DEGREES		
AREA UNDER CURVE UP TO 40.0 DEGREES	=	0.279 METRE RADIANS
AREA BETWEEN 30.0 AND 40.0 DEGREES	=	0.093 METRE RADIANS
MAXIMUM RIGHTING LEVER(GZ)	=	0.573 METRES
AND OCCURS AT 29.0 DEGREES		
MAXIMUM VERTICAL DEADWT MOMENT	=	19.8
ACTUAL VERTICAL DEADWT MOMENT	=	38.7

CONDITION NO. 5
FULLY LOADED ARRIVAL FROM CONDITION NO.4

STABILITY INFORMATION FOR TOWING STATE

K B ABOVE MLD. BASE = 0.861 METRES
HEIGHT OF TOW HOOK ABOVE BASELINE = 3.865 METRES

HEELING MOMENT = $T \times AD \times \cos(\text{ANGLE})$

WHERE T = ASSUMED 1/2 OF MAX. BOLLARD PULL
= 5.0 TONNES

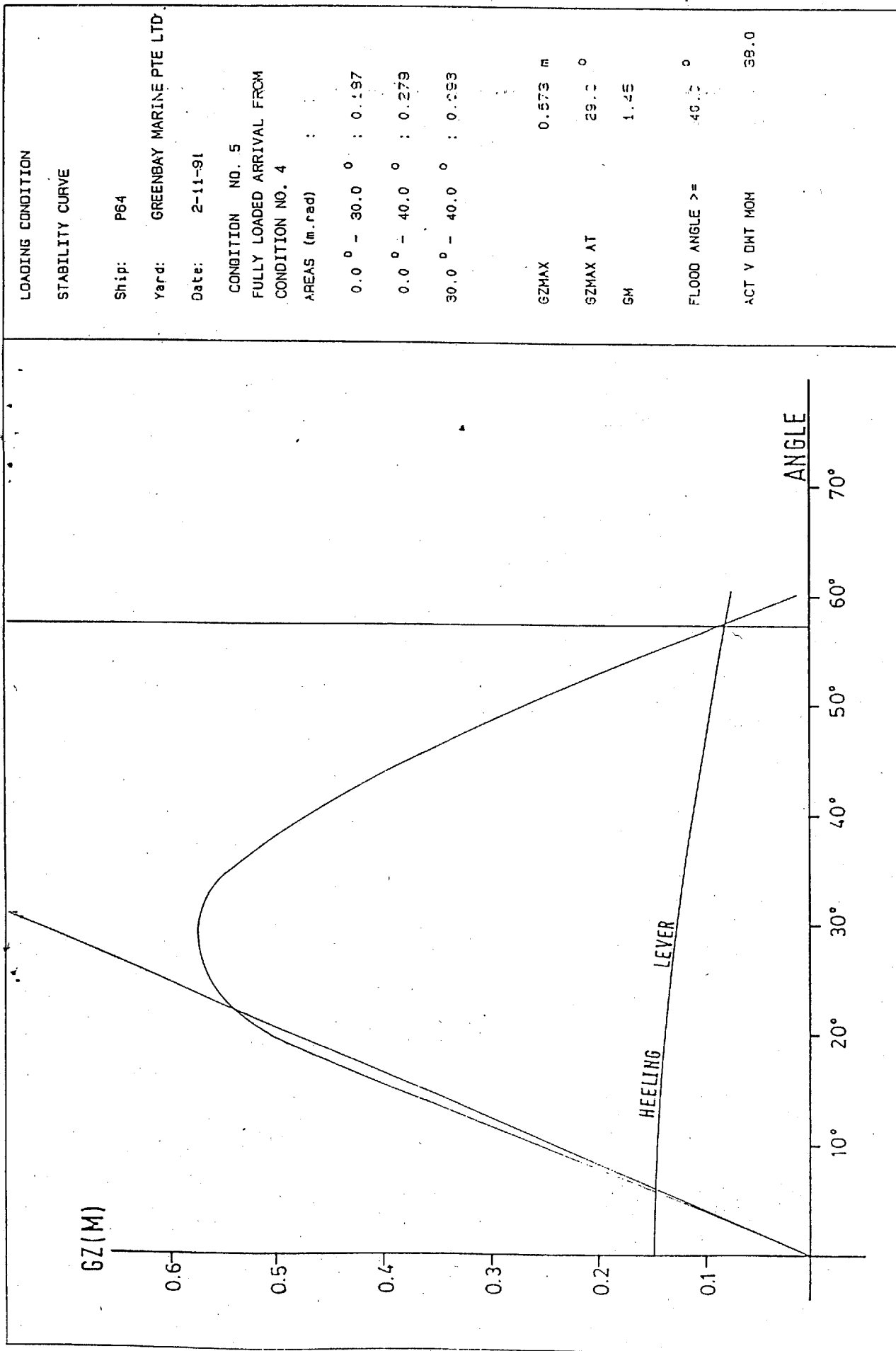
AD = VERTICAL HEIGHT OF TOWING POSITION
FROM KB
= 3.004 METRES

HEELING LEVER = HEELING MOMENT / DISPLACEMENT OF VESSEL

ANGLE OF HEEL DEGREES	$AD \times \cos(\text{ANGLE})$ METRES	HEELING MOMENT TONNE-METRES	HEELING LEVER METRES
0.000	3.004	15.020	0.148
10.000	2.958	14.792	0.145
20.000	2.823	14.114	0.139
30.000	2.602	13.008	0.128
40.000	2.301	11.506	0.113
50.000	1.931	9.655	0.095
60.000	1.502	7.510	0.074

THE AREA OF THE RESIDUAL DYNAMIC
STABILITY UP TO ANGLE OF HEEL OF
40.0 DEGREES PLUS THE ANGLE OF
FIRST INTERCEPT
(DOWNFLOODING ANGLE > 40 DEGREES)

= 0.220 METRE RADIANS



CONDITION NO. 6
FULL BALLAST DEPARTURE

ITEM	WEIGHT TONNES	V.C.G. ABOVE MLD. BASE METRES	VERT-L MOMENT ABOVE BASE TONNE METRES	L.C.G. FROM AP METRES	LONG-L MOMENT ABOUT THE AP TONNE METRES	FREE SURF. MOMENT TONNE METRES
CREW & EFFECTS	0.5	2.70	1.4	14.5	7.3	0.0
FUEL OIL DAY TK (P&S)	1.9	2.02	3.9	5.5	10.6	0.2
FUEL OIL TK NO.2 (P&S)	22.6	1.56	35.2	3.6	80.5	11.0
FRESH WATER TK (P&S)	11.4	2.28	26.0	16.6	188.8	0.7
F. W. HEADER TK (P&S)	1.5	3.01	4.6	7.5	11.4	0.1
FOREPEAK BALLAST TK (C)	17.2	2.32	39.9	19.2	329.5	13.4

DEADWEIGHT	55.1	2.01	110.8	11.4	628.0	25.4
LIGHTSHIP	88.8	2.43	216.2	10.0	884.8	0.0
DISPLACEMENT	143.9	2.27	327.1	10.5	1512.8	25.4

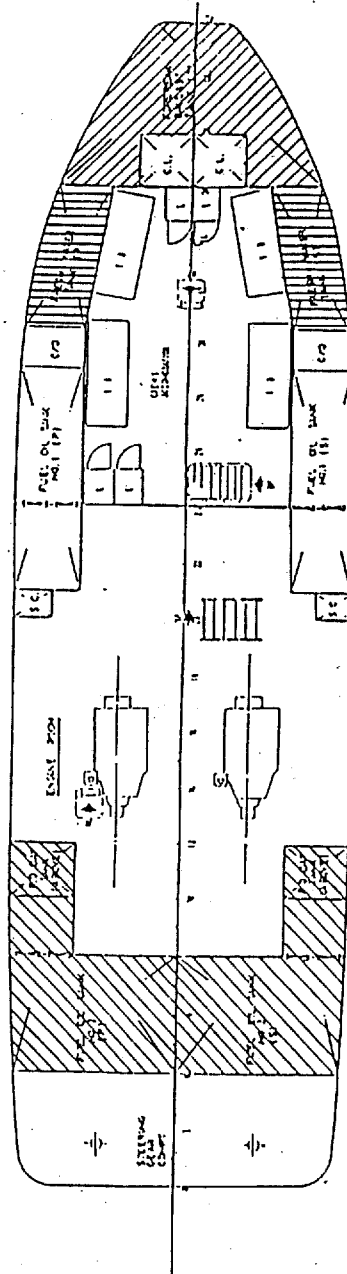
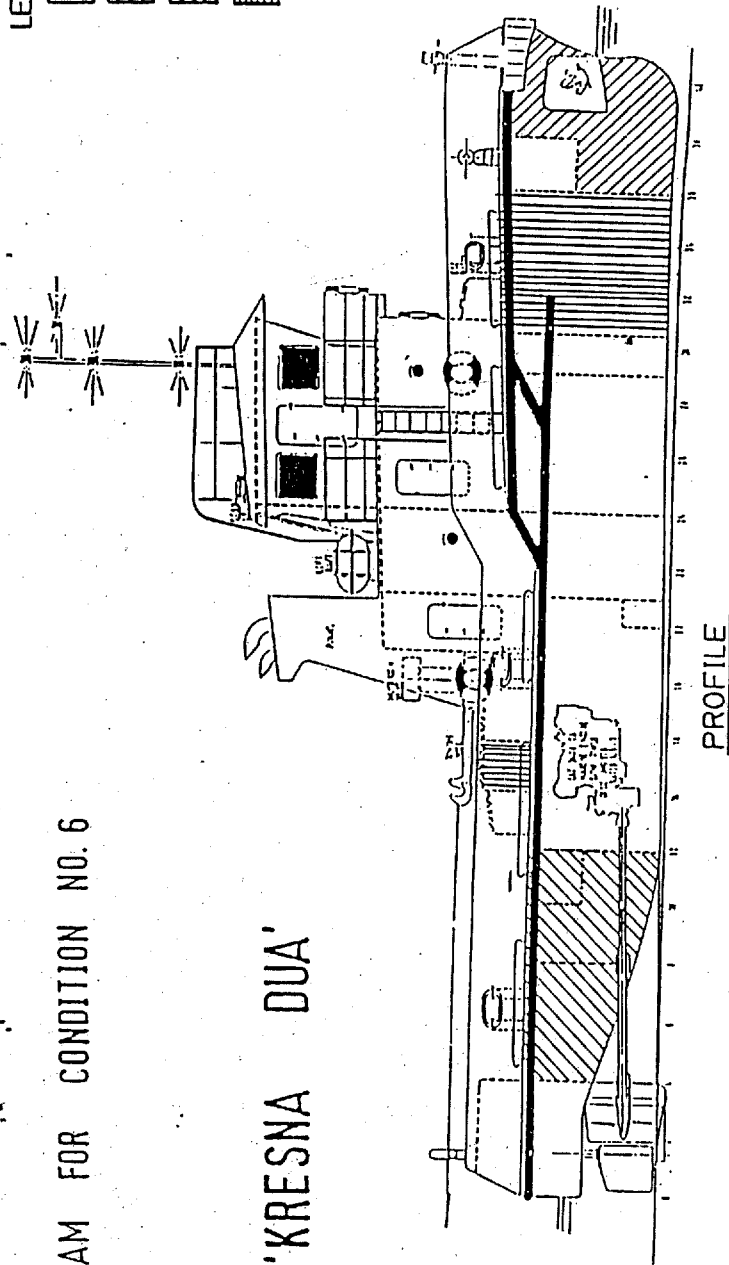
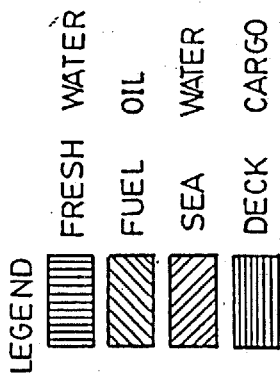
SAILING STATE

DENSITY OF WATER TAKEN AS 1.0252 (TONNES/CUBIC METRE)

FULL DRAFT AT L.C.F.	=	1.76 METRES
FULL DRAFT AFT AT MARKS	=	1.54 METRES
FULL DRAFT FWD AT MARKS	=	2.02 METRES
FULL MEAN DRAFT AT MIDSHIPS	=	1.79 METRES
TRIM BY BOW	=	0.61 METRES
K.M.T. ABOVE MLD. BASE	=	3.46 METRES
K.G. ABOVE MLD. BASE	=	2.27 METRES
G.M. (SOLID)	=	1.19 METRES
FREE SURFACE CORRECTION	=	0.18 METRES
G.M. (FLUID)	=	1.01 METRES

LOADING DIAGRAM FOR CONDITION NO. 6

'KRESNA DUA'



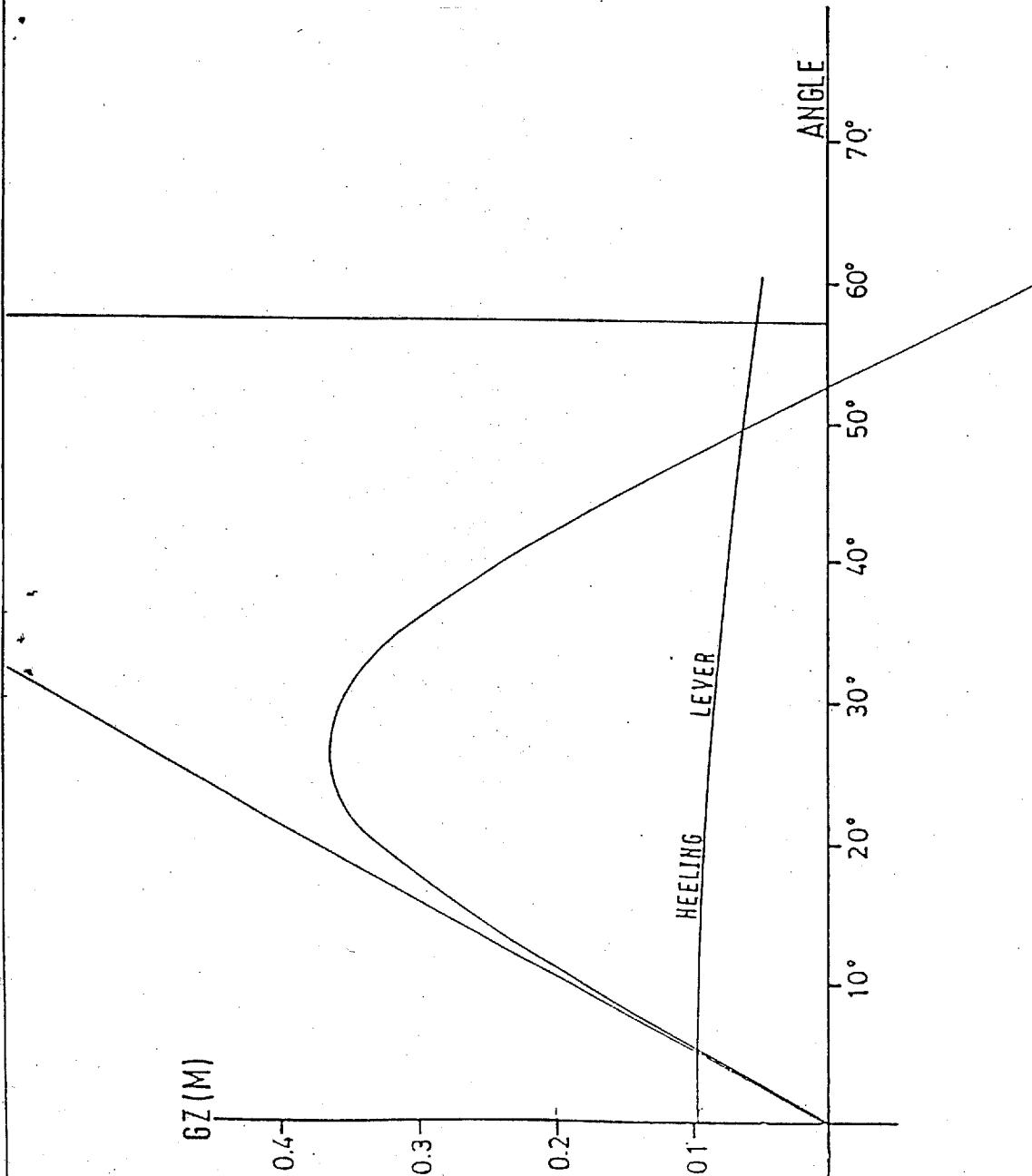
CONDITION NO. 6
FULL BALLAST DEPARTURE

STABILITY INFORMATION FOR SAILING STATE

K G ABOVE MLD. BASE	=	2.27 METRES
FREE SURFACE CORRECTION	=	0.18 METRES
KG CORRECTED (VIRTUAL KG)	=	2.45 METRES
MAXIMUM ALLOWABLE KG	=	2.11 METRES

ANGLE OF HEEL DEGREES	RIGHTING LEVER(KN) METRES	KG X SIN(ANGLE) METRES	RIGHTING LEVER(GZ) METRES
0.000	0.000	0.000	0.000
10.000	0.604	0.425	0.179
20.000	1.166	0.838	0.328
30.000	1.581	1.225	0.356
40.000	1.810	1.575	0.235
50.000	1.932	1.877	0.055
60.000	1.967	2.122	-0.154

AREA UNDER CURVE UP TO 30.0 DEGREES = 0.123 METRE RADIANS
 FLOOD ANGLE NOT LESS THAN 40.0 DEGREES
 AREA UNDER CURVE UP TO 40.0 DEGREES = 0.176 METRE RADIANS
 AREA BETWEEN 30.0 AND 40.0 DEGREES = 0.053 METRE RADIANS
 MAXIMUM RIGHTING LEVER(GZ) = 0.362 METRES
 AND OCCURS AT 27.0 DEGREES
 MAXIMUM VERTICAL DEADWT MOMENT = 87.2
 ACTUAL VERTICAL DEADWT MOMENT = 136.2



LOADING CONDITION

STABILITY CURVE

Ship: P64

Yard: GREENBAY MARINE PTE LTD.

Date: 2-11-91

CONDITION NO. 6

FULL BALLAST DEPARTURE

AREAS (m.rad) :

0.0 ° - 30.0 ° : 0.123

0.0 ° - 40.0 ° : 0.176

30.0 ° - 40.0 ° : 0.053

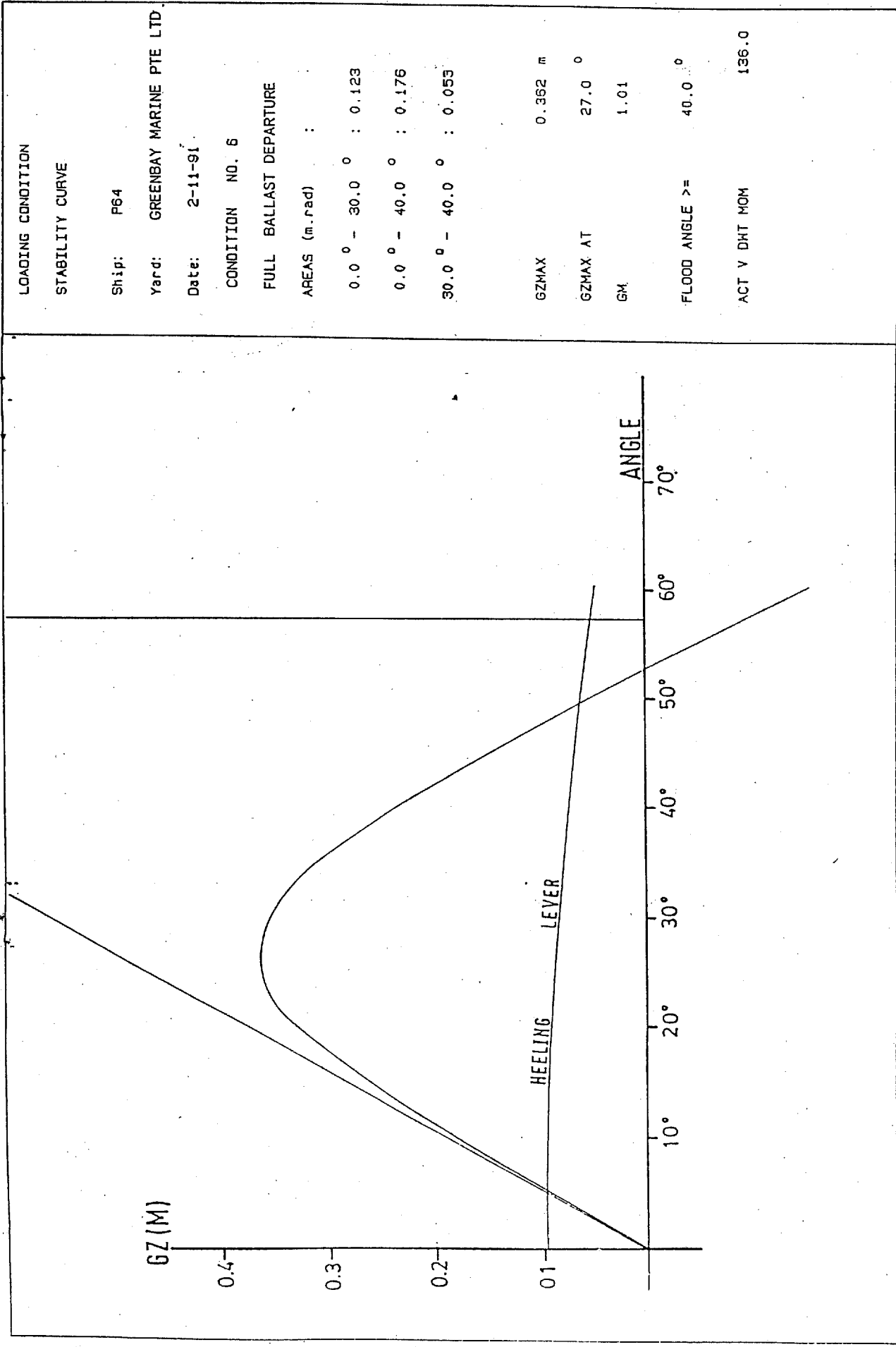
GZMAX 0.362 m

GZMAX AT 27.0 °

GM 1.01

FLOOD ANGLE >= 40.0 °

ACT V DWT MOM 136.0



CONDITION NO. 7

FULL BALLAST ARRIVAL FROM CONDITION NO.6

ITEM	WEIGHT TONNES	V.C.G. ABOVE MLD. BASE METRES	VERT-L MOMENT ABOVE BASE TONNE METRES	L.C.G. FROM AP METRES	LONG-L MOMENT ABOUT THE AP TONNE METRES	FREE SURF. MOMENT TONNE METRES
CREW & EFFECTS	0.5	2.70	1.4	14.5	7.3	0.0
FUEL OIL DAY TK (P&S)	1.9	2.02	3.9	5.5	10.6	0.2
FUEL OIL TK NO.2 (P&S)	2.3	0.72	1.6	4.4	9.9	11.0
FRESH WATER TK (P&S)	1.1	1.25	1.4	16.4	18.6	0.7
F. W. HEADER TK (P&S)	1.5	3.01	4.6	7.5	11.4	0.1
FOREPEAK BALLAST TK (C)	17.2	2.32	39.9	19.2	329.5	13.4

DEADWEIGHT	24.5	2.15	52.7	15.8	387.2	25.4
LIGHTSHIP	88.8	2.43	216.2	10.0	884.8	0.0
DISPLACEMENT	113.3	2.37	268.9	11.2	1272.0	25.4

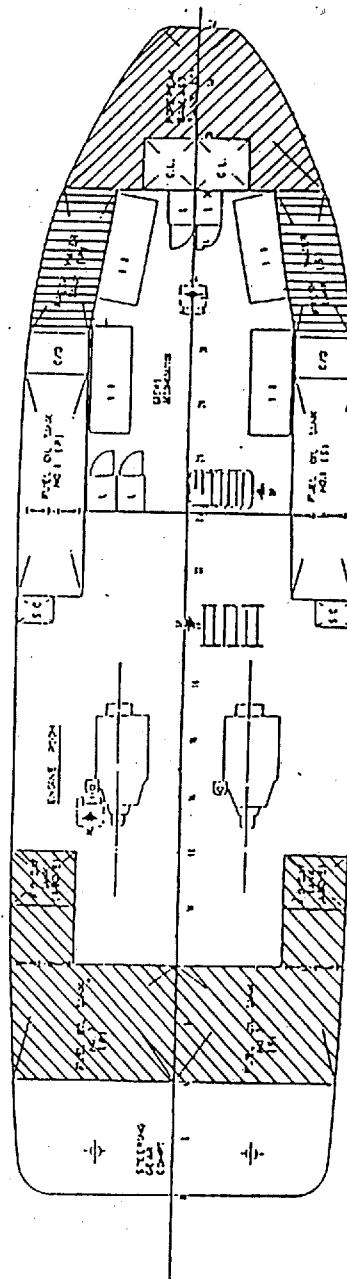
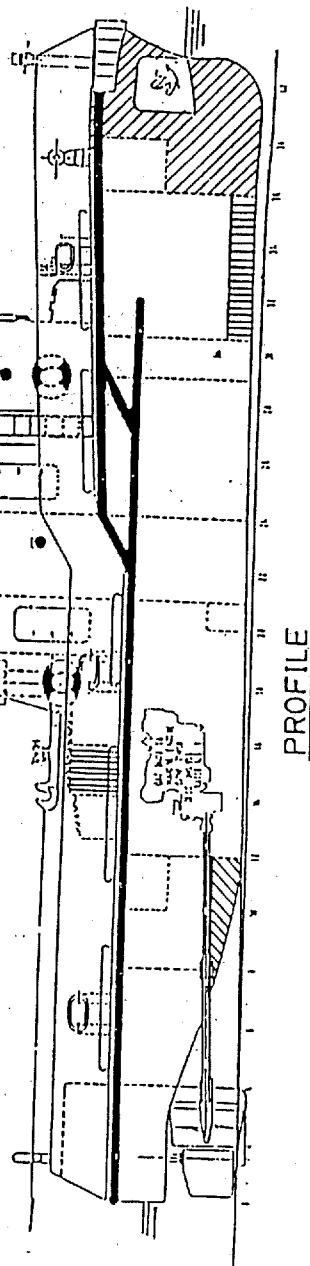
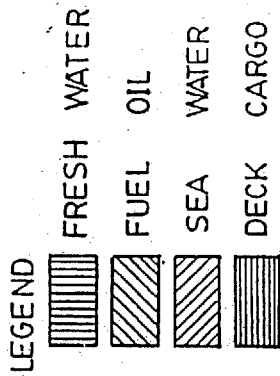
SAILING STATE

DENSITY OF WATER TAKEN AS 1.0252 (TONNES/CUBIC METRE)

FULL DRAFT AT L.C.F.	=	1.50 METRES
FULL DRAFT AFT AT MARKS	=	1.05 METRES
FULL DRAFT FWD AT MARKS	=	2.02 METRES
FULL MEAN DRAFT AT MIDSHIPS	=	1.53 METRES
TRIM BY BOW	=	0.97 METRES
K.M.T. ABOVE MLD. BASE	=	3.81 METRES
K.G. ABOVE MLD. BASE	=	2.37 METRES
G.M. (SOLID)	=	1.44 METRES
FREE SURFACE CORRECTION	=	0.22 METRES
G.M. (FLUID)	=	1.21 METRES

LOADING DIAGRAM FOR CONDITION NO.7

'KRESNA DUA'



CONDITION NO. 7

FULL BALLAST ARRIVAL FROM CONDITION NO.6

STABILITY INFORMATION FOR SAILING STATE

K G ABOVE MLD. BASE	=	2.37 METRES
FREE SURFACE CORRECTION	=	0.22 METRES
KG CORRECTED(VIRTUAL KG)	=	2.60 METRES
MAXIMUM ALLOWABLE KG	=	2.29 METRES

ANGLE OF HEEL DEGREES	RIGHTING LEVER(KN) METRES	KG X SIN(ANGLE) METRES	RIGHTING LEVER(GZ) METRES
0.000	0.000	0.000	0.000
10.000	0.660	0.451	0.209
20.000	1.312	0.888	0.423
30.000	1.765	1.299	0.467
40.000	2.014	1.670	0.344
50.000	2.122	1.990	0.132
60.000	2.129	2.249	-0.120

AREA UNDER CURVE UP TO 30.0 DEGREES	=	0.153 METRE RADIANS
FLOOD ANGLE NOT LESS THAN 40.0 DEGREES		
AREA UNDER CURVE UP TO 40.0 DEGREES	=	0.226 METRE RADIANS
AREA BETWEEN 30.0 AND 40.0 DEGREES	=	0.073 METRE RADIANS
MAXIMUM RIGHTING LEVER(GZ)	=	0.471 METRES
AND OCCURS AT 28.0 DEGREES		
MAXIMUM VERTICAL DEADWT MOMENT	=	43.4
ACTUAL VERTICAL DEADWT MOMENT	=	78.1

CONDITION NO. 7

FULL BALLAST ARRIVAL FROM CONDITION NO.6

STABILITY INFORMATION FOR TOWING STATE

K B ABOVE MLD. BASE = 0.925 METRES
 HEIGHT OF TOW HOOK ABOVE BASELINE = 3.865 METRES

HEELING MOMENT = $T \times AD \times \cos(\text{ANGLE})$

WHERE T = ASSUMED 1/2 OF MAX. BOLLARD PULL
 = 5.0 TONNES

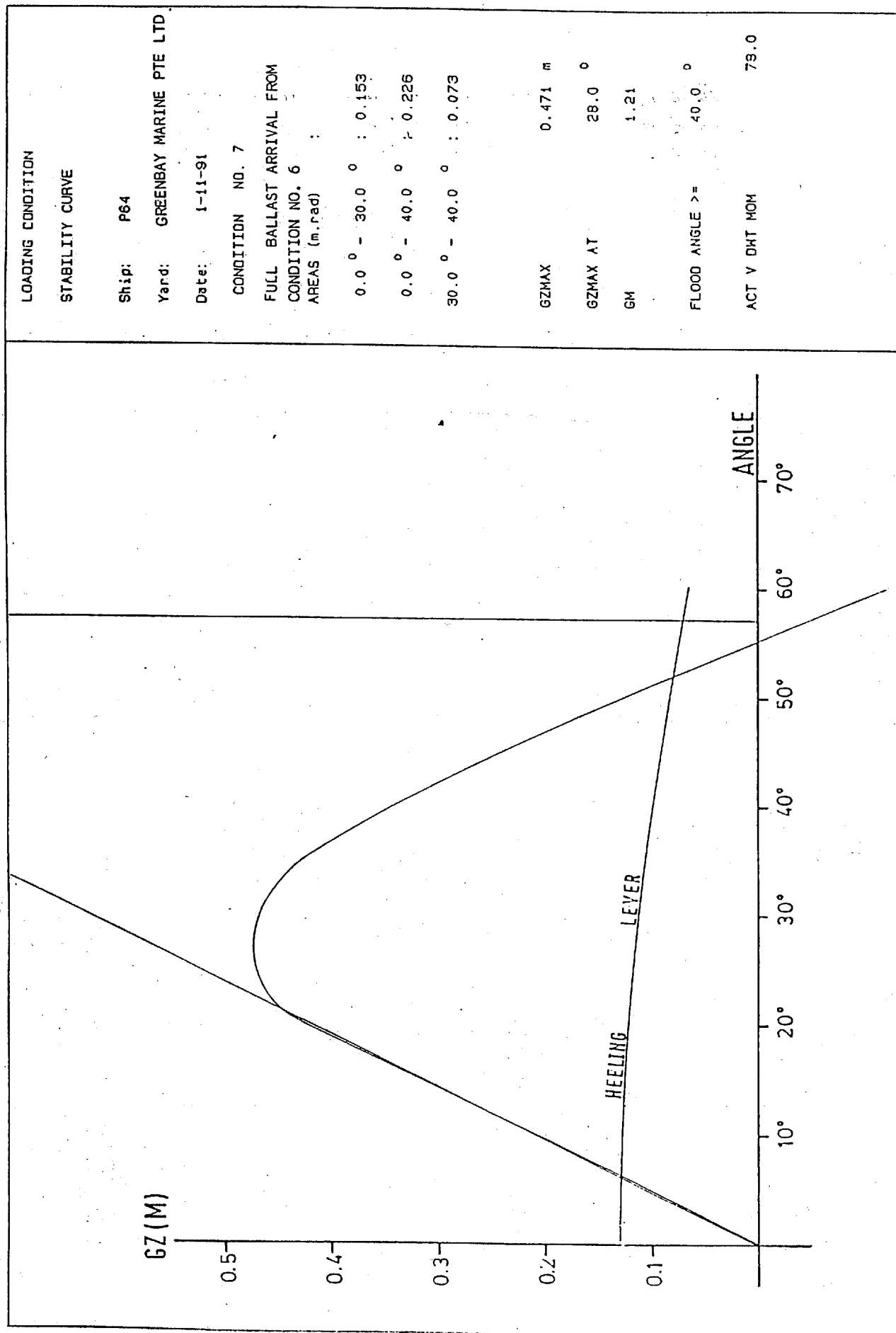
AD = VERTICAL HEIGHT OF TOWING POSITION
 FROM KB
 = 2.940 METRES

HEELING LEVER = $\text{HEELING MOMENT} / \text{DISPLACEMENT OF VESSEL}$

ANGLE OF HEEL DEGREES	AD x COS(ANGLE) METRES	HEELING MOMENT TONNE-METRES	HEELING LEVER METRES
0.000	2.940	14.700	0.130
10.000	2.895	14.477	0.128
20.000	2.763	13.813	0.122
30.000	2.546	12.731	0.112
40.000	2.252	11.261	0.099
50.000	1.890	9.449	0.083
60.000	1.470	7.350	0.065

THE AREA OF THE RESIDUAL DYNAMIC
 STABILITY UP TO ANGLE OF HEEL OF
 40.0 DEGREES PLUS THE ANGLE OF
 FIRST INTERCEPT
 (DOWNFLOODING ANGLE > 40 DEGREES)

= 0.171 METRE RADIANS



EXAMPLE SHOWING USE OF CROSS CURVES (KN)

The purpose of the Cross Curves is to enable Statical Stability Curves to be drawn for the ship in any sailing condition, e.g.:-

Assume the displacement of the ship to be 88.80 Tonnes, and the Vertical Center of Gravity 2.430 Meters above Baseline; as in Condition No. 1.

Then the Righting Lever $GZ = KN - KG \text{ Sine } \theta$;

where KN = Cross Curve ordinate;

KG = Corrected Vertical Center of Gravity above Base;

and θ = Angle of Heel

KN value	θ°	Sine θ	KG Sine θ	$GZ = KN - KG \text{ Sine } \theta$
0.730	10	0.174	0.423	0.307
1.427	20	0.342	0.833	0.594
1.888	30	0.500	1.217	0.671
2.143	40	0.643	1.565	0.578
2.253	50	0.766	1.865	0.388
2.249	60	0.866	2.109	0.140

Then by using the GZ values in the last column, a Statical Stability Curve can be drawn for the ship at that Displacement:-

